TamilPalam PRD

TamilPalam — Enterprise-Grade Product Requirements Document

Document Version: 1.0 **Status:** Final — Complete PRD **Prepared for:** Hackathon P10 — "How Can Tamil Be Taught to Tamil Children Living Abroad?" **Classification:** Research-Backed, Implementation-Ready

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1. Cover Page

Field	Detail
Product Name	TamilPalam (working name)
Meaning	"Tamil Bridge" — a bridge between English and Tamil
Product Category	Gamified Tamil language learning platform
Primary Audience	English-speaking children abroad learning Tamil
Document Type	Product Requirements Document (PRD)
Version	1.0 — Complete
Date	2025
Team Size	4 developers (hackathon context)
Hackathon Duration	15 hours
Target Deployment	Web (responsive), mobile-first PWA
Legal Notice	Name and domain availability not verified

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2. Document Control and Version History

Version	Date	Author	Changes
0.1	Day 0	Product Lead	Initial outline
0.5	Day 0	Full Team	Research integration
1.0	Day 0	Full Team	Complete PRD finalized

Document Owners: Product Lead (primary), Technical Lead (secondary)

Review Cycle: Post-hackathon, bi-weekly during active development

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3. Executive Summary

TamilPalam is a gamified, research-backed Tamil language learning platform designed for English-speaking children living outside Tamil Nadu who have little or no prior Tamil knowledge. The platform uses **Tanglish (Tamil written in English letters) as a temporary learning bridge**, progressively reducing transliteration support as learners gain mastery of Tamil script, vocabulary, sentences, and reading comprehension.

The product addresses a critical gap: existing Tamil learning apps for children abroad either assume prior script knowledge, lack progressive transliteration reduction, or provide insufficient audio-first instruction. TamilPalam differentiates through its **Progressive Tanglish Reduction** system — a structured five-stage pathway that moves learners from full Tanglish support to independent Tamil reading.

Core deliverables (P0): English/Tamil interface localization, onboarding and assessment, level-based curriculum (Levels 0–7), letter/word/sentence learning, listening and speaking practice, quizzes, persistent progress, gamification, and responsive design.

Differentiating features (P1): Progressive Tanglish Reduction, personalized revision, real-life Tamil missions, audio-first learning, adaptive recommendations, and cultural storytelling.

Future expansion (P2): Tamil Research Assistant for literary content explanation.

The MVP is scoped for a 4-person student team with AI-assisted development tools and a 15-hour hackathon deadline. The recommended stack is **React + TypeScript + Vite + Tailwind CSS** on the frontend, **Supabase (PostgreSQL + Auth + Storage + RLS)** on the backend, with no custom API server unless justified.

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4. Product Vision

Build a polished, interactive, gamified Tamil learning platform that helps an English-speaking beginner progress from recognizing Tamil letters to understanding words, reading sentences, participating in conversations, and reading simple paragraphs and stories.

Tanglish is the initial bridge between English and Tamil. The product must **not stop at teaching Tanglish spellings**. Its educational objective is to progressively reduce dependence on Tanglish and help learners recognize and read Tamil script independently.

The experience should feel like a thoughtfully designed educational product created by a premium product-design team — not a generic AI-generated website.

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5. Problem Statement

P10 — How Can Tamil Be Taught to Tamil Children Living Abroad?

Children living abroad who do not speak Tamil at home need an engaging online way to learn Tamil and continue learning consistently. The learning platform should encourage their interest and measure their learning progress.

The solution must specifically address:

1. How to teach Tamil to children who have little or no prior Tamil knowledge.
2. How to make learning interesting and sustainable.
3. How to measure learning progress.
4. How to help learners gradually develop real Tamil reading, writing, listening, and speaking skills.

The learner knows English and is learning Tamil. The problem is **not** teaching English to Tamil speakers.

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6. Problem Validation and Supporting Evidence

Evidence Area	Finding	Source Type
Heritage language attrition	Children who actively engage with heritage language experience academic success, stronger socio-emotional skills, and enhanced well-being	Peer-reviewed research (Taylor & Francis, 2025)
Transliteration dependence	Existing apps show Tamil script alongside transliteration but lack systematic reduction	Competitor analysis (Google Play, App Store)
Gamification effectiveness	Streaks, XP, and badges significantly predict continued app usage and motivation	IEEE, Zenodo, Lund University studies
Pronunciation assessment limitations	Tamil speech recognition achieves WER of 0.55–0.71 on dialectal speech; children's speech performs worse	ACL Anthology, Semantic Scholar
Privacy regulation landscape	COPPA 2.0 (2025) expands definition of personal information; GDPR-K requires verifiable parental consent	FTC, Cookie-Script analysis

Assumption requiring validation: The specific efficacy of progressive Tanglish reduction as a pedagogical method has not been independently validated. A pilot study is recommended before making educational efficacy claims.

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7. Product Goals

Goal ID	Goal	Success Metric	Priority
G-001	Enable English-speaking children to read Tamil script independently	70% of active learners reach Stage D (audio-only support) within 60 sessions	P0
G-002	Sustain learning engagement through gamification	40% weekly return rate among onboarded learners	P0
G-003	Provide measurable learning progress	Dashboard shows skill mastery per level with 95% data completeness	P0
G-004	Support both English and Tamil interface languages fully	100% of UI strings localized; zero missing translation warnings in production	P0
G-005	Maintain child safety and privacy compliance	Zero unauthorised access to learner data; parental consent flow operational	P0

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8. Non-Goals

- Teaching English to Tamil speakers.
- Claiming fluency within a fixed number of days.
- Replacing formal Tamil education or teacher-led instruction.
- Providing certified language proficiency assessments.
- Supporting languages other than Tamil (in MVP).
- Building a social network or chat feature between learners.
- Monetization targeting children (subscriptions, ads, in-app purchases for the MVP).
- Full Tamil Research Assistant module in the MVP.

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9. Target Users

Primary Persona: “Aarav”

Attribute	Detail
Name	Aarav
Age	7–12 years
Location	United States / United Kingdom / Australia / Singapore
Languages	English (primary), hears Tamil from parents/grandparents
Tamil Level	Little to no reading; may understand some spoken words
Device	Tablet (iPad/Android), sometimes shared family laptop
Pain Points	Cannot read Tamil script; finds traditional Tamil classes boring; feels disconnected from heritage
Motivations	Wants to understand what grandparents say; wants to read Tamil storybooks; enjoys games and rewards
Tech Comfort	High — uses educational apps at school

Secondary Personas

Persona	Description	Needs
Parent (Priya)	Wants child to maintain Tamil connection	Progress visibility, low time commitment, safe platform
Teacher (Murugan)	Weekend Tamil school instructor	Classroom-friendly content, printable progress reports
Heritage Learner (Divya)	16-year-old who understands spoken Tamil but cannot read	Faster progression, adult-appropriate content tone
Adult Beginner (Raj)	35-year-old spouse learning Tamil	Self-paced, no childish imagery,

		practical vocabulary

10. User Personas (Expanded)

Persona 1: Aarav — The Curious Child

- **Goals:** Read Tamil like his grandmother; play games; earn badges.
- **Frustrations:** Tamil looks like "squiggles"; adults speak too fast.
- **Quote:** "I want to surprise Paati by reading her a story."
- **Design implication:** Visual, audio-heavy, short sessions (5–10 min), celebratory feedback.

Persona 2: Priya — The Busy Parent

- **Goals:** Ensure child learns Tamil; track progress without hovering.
- **Frustrations:** No time to teach; existing apps are too complex.
- **Quote:** "I just want to know he's actually learning, not just tapping screens."
- **Design implication:** Parent dashboard with weekly summary; email reports.

Persona 3: Murugan — The Weekend Teacher

- **Goals:** Supplement classroom learning; assign homework.
- **Frustrations:** Students forget between weekend classes.
- **Quote:** "If they practice during the week, we can move faster on Saturdays."
- **Design implication:** Shareable progress reports; assignment features (P1).

Persona 4: Divya — The Heritage Learner

- **Goals:** Learn to read Tamil script; understand written Tamil.
- **Frustrations:** Apps for kids feel babyish; adult courses are too slow.
- **Quote:** "I understand what Amma says, but I can't read her WhatsApp messages."
- **Design implication:** Tone-neutral interface; skip option for oral vocabulary; script-focused pathway.

Persona 5: Raj — The Adult Beginner

- **Goals:** Learn Tamil to communicate with in-laws; travel confidence.
- **Frustrations:** No structured self-study path; pronunciation anxiety.
- **Quote:** "I don't need to write poetry. I just want to have a conversation."
- **Design implication:** Practical phrase-first approach; optional script learning.

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11. Jobs to Be Done

Job ID	When I…	I want to…	So I can…
JTBD-01	See Tamil text	Recognize at least some letters	Feel less intimidated
JTBD-02	Hear my grandmother speak	Understand basic words	Respond appropriately
JTBD-03	Have 10 minutes free	Complete a short lesson	Feel productive
JTBD-04	Finish a week of learning	See my progress	Stay motivated
JTBD-05	Struggle with a word	Get a hint without feeling stupid	Keep trying
JTBD-06	Master a level	Earn recognition	Feel proud
JTBD-07	Want to read Tamil	Transition from Tanglish to script	Achieve independence

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12. User Pain Points

Pain Point	Frequency	Severity	Current Workaround
Tamil script looks unfamiliar and intimidating	Every session	High	Avoid script; rely on Tanglish
Pronunciation uncertainty (retroflex,	Every lesson	High	Guess or skip speaking

long vowels)			
Lessons too long / attention span	Every session	Medium	Abandon app
No visible progress	Weekly	Medium	Lose motivation
Content feels too childish (older learners)	Every session	Medium	Stop using app
No offline access	Travel/commute	Medium	Skip learning days
Tanglish never goes away	Long-term	High	Never learn real script

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13. User Stories

ID	As a…	I want to…	So that…	Priority
US-001	Child learner	Choose English or Tamil for the app	I understand everything	P0
US-002	Child learner	Take a quick assessment	Start at my level	P0
US-003	Child learner	See Tamil letters with audio	Learn pronunciation	P0
US-004	Child learner	Practice with Tanglish support	Not feel overwhelmed	P0
US-005	Child learner	Earn streaks and badges	Stay motivated	P0
US-006	Child learner	Resume a lesson after interruption	Not lose progress	P0
US-007	Child learner	See Tanglish gradually reduce	Learn real Tamil	P1
US-008	Parent	View child's progress	Know they are learning	P1
US-009	Teacher	Assign lessons to students	Supplement class	P2
US-010	Adult learner	Skip childish content	Learn at my level	P1

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14. Product Scope

A. Core Product — Mandatory (P0)

1. English and Tamil interface localization
2. Learner onboarding and initial skill assessment
3. Level-based Tamil learning curriculum (Levels 0–7)
4. Tamil letters and pronunciation
5. Tamil words and vocabulary
6. Basic sentences and sentence construction
7. Listening and speaking practice
8. Reading practice and short paragraphs
9. Interactive quizzes and educational games
10. Tanglish-assisted learning
11. Lesson completion and assessment results
12. Learning progress dashboard
13. Streaks, points, badges, and appropriate rewards
14. Persistent data storage
15. Responsive design for mobile, tablet, and desktop

B. High-Value Differentiating Features (P1)

1. Progressive Tanglish Reduction
2. Personalised Revision
3. Real-Life Tamil Missions
4. Audio-First Learning
5. Adaptive Learning Recommendations
6. Structured transition from transliteration to native Tamil script
7. Cultural storytelling integrated into language learning

C. Future Expansion (P2)

- **Tamil Research Assistant:** Literary content explanation, Thirukkural study, source verification, uncertainty communication, hallucination prevention

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15. MVP Prioritisation

Priority	Definition	Features
P0	Essential for working prototype	Localization, onboarding, Levels 1–4 curriculum, letter/word/sentence learning, basic quizzes, audio playback, progress persistence, gamification (streaks/points/badges), responsive design, Tanglish support
P1	Important enhancement	Progressive Tanglish Reduction, personalized revision, real-life missions, adaptive recommendations, parent dashboard, cultural storytelling
P2	Future expansion	Tamil Research Assistant, teacher assignments, advanced speech assessment, offline mode, multiple dialect support

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16. Competitive Research

Competitor	Target Audience	Key Features	Transliteration	Strengths	Weaknesses	TamilPalam Differentiation
Kinder Tamil	Children 3–8	Play-based lessons, Tamil culture, 200+ words, podcasts, offline	Not verified	Child-friendly design, cultural integration	Limited vocabulary, no progressive reduction	Progressive Tanglish Reduction, structured levels
Tamil Agaram Pro	Children 4–10	1800+ words, animations, voice-overs, games, quizzes, progress tracking	Phonetic pronunciations shown	Extensive vocabulary, parent dashboard	No transliteration reduction strategy	Audio-first design, five-stage reduction
Learn Tamil through English	Beginners	Script + transliteration + meaning, quizzes	Yes (static)	Clear triple presentation	No gamification depth, no progression	Gamified, adaptive progression
Bhasha: Learn Indian Languages	Adults/teens	Gamified levels, drag-drop, MCQ, accuracy adaptation	Not verified	Multi-language support	Not Tamil-focused, no script-first approach	Tamil-specific depth
Duolingo (reference)	General	XP, streaks, leaderboards, gamification	N/A	Proven engagement mechanics	Not available for Tamil	Tamil-specific curriculum, child-safe gamification

Verified finding: No existing Tamil learning app combines progressive transliteration reduction with gamified, level-based curriculum. TamilPalam addresses this gap.

Assumption requiring validation: Competitor pricing and feature availability may change. Verify before making public claims.

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17. Product Positioning

TamilPalam positions itself as the **premier Tamil learning platform for English-speaking children abroad** — not a generic language app, not a Tamil-English dictionary, and not a formal classroom replacement. It occupies the space between “fun but shallow” children’s Tamil apps and “rigorous but inaccessible” formal Tamil education.

Positioning statement: For English-speaking children abroad who want to connect with their Tamil heritage, TamilPalam is a gamified learning platform that uses Tanglish as a bridge to build real Tamil reading and comprehension skills. Unlike other apps, TamilPalam progressively reduces transliteration support so learners achieve independent Tamil script literacy.

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18. Differentiation Strategy

Differentiator	Description	Competitive Gap	Defensibility
Progressive Tanglish Reduction	Five-stage system reducing transliteration as mastery grows	No competitor implements this	High — requires pedagogical design and data
Audio-First Learning	Pronunciation before writing; listen-speak-read sequence	Most apps are text-first	Medium
Child-Safe Gamification	Streaks and badges without punitive loss or leaderboards	Duolingo has streak loss anxiety	Medium — ethical design
Real-Life Missions	Practical tasks (call grandparent, read menu)	None in Tamil learning	High
Cultural Storytelling	Tamil stories integrated into lessons	Limited in competitors	Medium

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19. Learning Principles

Principle	Description	Application
Audio-first	Learners hear correct pronunciation before seeing text	Every new letter/word begins with audio
Progressive disclosure	Tanglish support reduces as mastery increases	Five-stage system
Spaced repetition	Review intervals expand with successful recall	Revision queue algorithm
Interleaving	Mix related concepts in practice	Lessons blend letters + vocabulary + sentences
Active recall	Learners produce answers, not just recognize	Speaking, typing, matching exercises
Immediate feedback	Correct/incorrect feedback within 1 second	Quiz response handling
Low-stakes failure	Mistakes are learning opportunities, not punishments	No streak loss for wrong answers
Cultural relevance	Content connects to Tamil culture and family life	Stories, missions, proverbs

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20. Curriculum Framework

CEFR Alignment

TamilPalam Level	CEFR Equivalent	Can-Do Descriptor
Level 0	Pre-A1	Recognize Tamil script exists; understand assessment
Level 1	Pre-A1	Recognize all vowels and basic consonants with audio support
Level 2	A1 (receptive)	Understand 50+ spoken words with visual context
Level 3	A1	Read simple 2–3 syllable words in Tamil script
Level 4	A1+	Construct simple sentences about self and family
Level 5	A2 (listening)	Follow short guided dialogues
Level 6	A2	Read short paragraphs with comprehension
Level 7	A2+	Participate in real-world Tamil interactions

Level Structure

Every level contains:

- **Learning objectives** (3–5 per level)
- **Entry requirements** (mastery of previous level or assessment score)
- **Skills taught** (reading, writing, listening, speaking, vocabulary, grammar)
- **Example lessons** (5–8 per level)
- **Lesson sequence** (prerequisite-ordered)
- **Activities** (matching, MCQ, fill-blank, speaking, listening, sentence-building)
- **Assessment types** (formative quiz, unit test, speaking check)

- **Pass/mastery criteria** (80% accuracy on unit test)
- **Remediation paths** (targeted revision for failed items)
- **Completion criteria** (all lessons + unit test passed)
- **Estimated time** (Level 0: 10 min; Levels 1–7: 45–90 min each)
- **Tanglish policy** (see Section 11)
- **Audio requirements** (prerecorded for all letters, words, sentences)
- **Accessibility requirements** (WCAG 2.1 AA, screen reader labels)

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21. Tamil-Script Learning Model

Unicode Foundation

Tamil script in Unicode occupies code points U+0B80–U+0BFF. Independent vowels (Uyir) occupy U+0B85–U+0B94; consonants (Mei) occupy U+0B95–U+0BB9. Vowel signs (Uyirmei) are combining characters that modify consonants. The virama (U+0BCD, ்) suppresses the inherent vowel.

Critical implementation note: Tamil is a complex script requiring proper rendering support. Browsers and OS font stacks must handle combining characters, ligatures (e.g., கூ), and the split vowel sign ொ (U+0BCA). Use Unicode NFC normalization for all stored and displayed text. Test rendering in Chrome, Safari, Firefox, and Edge on Windows, macOS, iOS, and Android.

Letter Introduction Sequence

Phase	Content	Audio	Tanglish	Visual
1	அ, ஆ, இ, ஈ, உ, ஊ	Yes	Yes	Animated letter formation
2	எ, ஏ, ஐ, ஒ, ஓ, ஔ	Yes	Yes	Letter + example word
3	க, ங, ச, ஞ, ட, ண	Yes	Yes	Letter + sound + image
4	த, ந, ப, ம, ய, ர, ல, வ, ழ, ள, ற, ன	Yes	Yes	Retroflex distinction highlighted
5	Combined forms (க் + vowel signs)	Yes	Reduced	Formation animation

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22. Tanglish Transliteration Model

The Challenge

Tamil sounds do not map perfectly to English letters. Key issues:

- **Long vs. short vowels:** ஆ (ā) vs. அ (a) — Tanglish "aa" vs "a"
- **Retroflex consonants:** ள (ḷ), ழ (ḻ), ண (ṇ) — no exact English equivalent
- **Voiced/unvoiced distinction:** க can be "ka" or "ga" depending on position
- **Dental vs. retroflex:** த (dental "th") vs. ட (retroflex "t")

Transliteration Rules

Tamil	IPA	Tanglish (Primary)	Alternative	Note
அ	/a/	a	—	Short
ஆ	/aː/	aa	a	Long
இ	/i/	i	—	Short
ஈ	/iː/	ee	i	Long
க	/k/	ka	ga	Context-dependent voicing
ங	/ŋ/	nga	—	Velar nasal
ச	/t͡ʃ/	cha	sa	Context-dependent
ஞ	/ɲ/	nya	—	Palatal nasal
ட	/ʈ/	ta	da	Retroflex
ண	/ɳ/	na	—	Retroflex nasal

த	/t̪/	tha	da	Dental
ந	/n̪/	na	—	Dental nasal
ள	/ɭ/	la	—	Retroflex lateral
ழ	/ɻ/	zha	l	Approximant
ற	/r/	ra	—	Trill
ன	/n/	na	—	Alveolar nasal

Handling ambiguity: Where multiple Tanglish spellings are common (e.g., "zha" vs "l" for ழ), the platform uses one primary spelling and displays alternatives on request. The audio is the authoritative pronunciation reference.

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23. Progressive Tanglish Reduction

Five-Stage System

Stage	Tamil Script	Tanglish	English Meaning	Audio	Trigger to Advance
A	Always shown	Always shown	Always shown	Auto-play	80% accuracy on 10 items
B	Always shown	Always shown	On demand	Auto-play	85% accuracy on 15 items
C	Always shown	On demand	Hidden	On demand	85% accuracy on 15 items + 3 sessions
D	Always shown	Hidden	Hidden	On demand	90% accuracy on 20 items
E	Always shown	Hidden	Hidden	Hidden	Mastery confirmed

Readiness Evidence

- Stage advancement is based on **demonstrated accuracy across multiple item types** (recognition, recall, production).
- Performance is tracked per skill (letter recognition, word reading, sentence comprehension).
- A decrease in performance triggers **automatic regression** to the previous stage for affected items only.
- Learners can **manually request Tanglish support** at any stage via a persistent "Need help?" button.
- Accessibility support (larger text, audio repeat, screen reader labels) is **never removed**.

Mastery Calculation

```
skill_mastery = (correct_recalls / total_attempts) × recency_weight × difficulty_modifier
```

- `recency_weight`: Recent attempts weighted higher (exponential decay over 14 days)
- `difficulty_modifier`: Harder items contribute more to mastery score
- Mastery threshold per skill: 0.80 for stage advancement, 0.90 for level completion

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24. Vocabulary Learning

Categories

Category	Example Words	Level	Audio	Visual
Family	அம்மா (amma), அப்பா (appa), அக்கா (akka)	2	Yes	Family illustration
Greetings	வணக்கம் (vanakkam), நன்றி (nandri)	2	Yes	Greeting gesture
Numbers	ஒன்று (ondru), இரண்டு (irandu)	2	Yes	Number + objects
Colours	சிவப்பு (sivappu), பச்சை (pachai)	2	Yes	Colour swatch
Food	சோறு (soru), தண்ணீர் (thanneer)	2	Yes	Food image
Animals	நாய் (naai), பூனை (poona)	2	Yes	Animal image
Actions	வா (vaa), போ (po)	3	Yes	Action animation

Spaced Repetition Algorithm

Interval	Trigger

1 day	Initial learning
3 days	First successful recall
7 days	Second successful recall
14 days	Third successful recall
30 days	Fourth successful recall
60 days	Mastered

Failed recall resets interval to 1 day for that item.

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25. Sentence Construction

Progression

Level	Sentence Type	Example	Support
3	2-word phrases	நல்ல பையன் (nalla paiyan) — good boy	Full Tanglish
4	3–4 word sentences	என் பெயர் ஆரவ் (en peyar Aarav) — my name is Aarav	Tanglish on demand
4	Questions	உனக்கு எப்படி இருக்கு? (unakku eppadi irukku?) — how are you?	Audio + Tanglish
5	Connected sentences	எனக்கு பழம் பிடிக்கும். ஆனால் காய்கறி பிடிக்காது. — I like fruit. But I don't like vegetables.	Audio + minimal Tanglish
6	Paragraph	Short descriptive paragraph	Audio only

Sentence-Building Exercise Types

1. **Word ordering:** Drag words to form a correct sentence.
2. **Fill in the blank:** Choose the correct word from options.
3. **Translation:** Given English, select/type Tamil.
4. **Listening dictation:** Hear Tamil, type/select words.

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26. Listening Practice

Audio Types

Type	Description	Speed Control	Repeat
Letter pronunciation	Single letter sound	0.5x, 0.75x, 1x	Yes
Word pronunciation	Full word with stress	0.5x, 0.75x, 1x	Yes
Sentence audio	Full sentence, natural speed	0.75x, 1x	Yes
Dialogue	Two-speaker conversation	0.75x, 1x	Yes
Story	Narrative passage	0.75x, 1x	Yes

Exercise Flow

1. Audio plays automatically (once).
2. Learner listens and selects from 3–4 options.
3. If incorrect, audio repeats with highlighted segment.
4. If correct, visual confirmation + next item.

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27. Speaking Practice

Scope Limitation

Important: Phoneme-level pronunciation assessment for Tamil is not commercially reliable at MVP scale. Tamil speech recognition models achieve WER of 0.55–0.71 on dialectal speech, and children's speech performs worse. Therefore:

- **P0:** Speaking practice uses **record-and-compare** (learner records, hears their voice alongside model audio) without AI scoring.
- **P1:** Optional speech-to-text transcription via Web Speech API or third-party service, presented as "speech recognition" not "pronunciation assessment."
- **P2:** Phoneme-level assessment if a validated Tamil assessment API becomes available.

Recording Flow

1. Model audio plays.
2. Learner taps "Record."
3. Microphone permission requested (with plain-language explanation).
4. Recording captured (max 10 seconds).
5. Playback: learner's voice → model audio.
6. Self-assessment prompt: "How close was your pronunciation?"
7. Recording deleted from server after 24 hours (or immediately if not saved).

Consent and Privacy

- Microphone access requires explicit consent.
- For users under 13, parental consent required before any recording.
- Recordings are never shared, used for training, or retained beyond 24 hours.
- Recording feature can be disabled by parent/guardian in settings.

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28. Reading Practice

Progression

Level	Text Type	Length	Tanglish Support	Comprehension
3	Single words	1 word	Full	Picture match
4	Phrases	2–4 words	On demand	Select meaning
5	Simple sentences	3–6 words	Reduced	True/false
6	Paragraphs	20–40 words	Hidden	MCQ
7	Short stories	50–100 words	Hidden	Open response

Reading Confidence Measurement

Reading confidence is measured through:

- **Accuracy:** Percentage of correctly read words (via self-report or audio match)
- **Fluency:** Time to read passage (logged, not scored)
- **Comprehension:** MCQ accuracy on passage content
- **Self-report:** "How easy was this to read?" (1–5 scale)

Limitation: Self-reported confidence is subjective and not equivalent to proficiency. The platform presents these metrics as learning indicators, not certified assessments.

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29. Story-Based Learning

Story Types

Type	Level	Length	Audio	Tanglish
Picture story	2–3	4 panels	Yes	Full
Short narrative	4–5	50 words	Yes	Reduced
Cultural story	5–6	100 words	Yes	Optional

Folk tale	6–7	200 words	Yes	Hidden

Cultural Story Examples

- “The Clever Crow” (panchatantra-style)
- “The Festival of Pongal”
- “Grandmother’s Kitchen”

Each story includes vocabulary preview, reading, comprehension questions, and optional speaking prompt (“Tell the story in your own words”).

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30. Gamification

Mechanics

Mechanic	Description	Rules	Ethical Guardrail
Daily Goal	Target lessons per day	Default: 1 lesson. Adjustable 1–3.	No penalty for missing
Streak	Consecutive days of learning	1 lesson = streak day. Streak freeze available (1 per week).	No streak loss for technical errors. Grace period: 1 day.
Experience Points (XP)	Points for completing activities	10 XP per correct answer, 5 XP per lesson completion, 50 XP per level completion	XP correlates with learning effort, not time spent
Badges	Skill-based achievements	“First Letter,” “10 Words,” “First Sentence,” “Week Warrior”	Badges tied to demonstrated skills
Level Completion	Unlock next level	80% mastery on unit test	No pay-to-unlock
Milestones	Personal progress markers	“50 Words Learned,” “First Story Read”	No comparison to others

Prohibited Gamification Patterns

- **No punitive streak loss** for technical errors or verified sickness.
- **No public leaderboards** exposing children’s names or scores.
- **No random reward boxes** (loot box mechanics).
- **No manipulative notifications** (e.g., “Your friend is ahead of you!”).
- **No monetization of children’s data.**

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31. Rewards and Achievements

Badge Categories

Category	Badge	Criteria	Icon
Letters	அ Explorer	Recognize all vowels	Tamil letter அ
Letters	Mei Master	Recognize 10 consonants	Tamil letter �க
Vocabulary	Word Collector	Learn 25 words	Book stack
Vocabulary	Word Master	Learn 100 words	Trophy with book
Reading	First Reader	Read first word	Eye + book
Reading	Story Reader	Complete first story	Open book
Streak	Week Warrior	7-day streak	Flame with 7

Streak	Month Master	30-day streak	Flame with 30
Mastery	Level Up	Complete Level 1	Star with 1
Mastery	Script Independent	Reach Stage E	Tamil letter with checkmark

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32. Personalisation

Adaptive Recommendations

Signal	Recommendation
Struggles with retroflex consonants	Extra listening practice for ள, ழ, ண
Fast completion of vocabulary	Skip to sentence construction
Low speaking participation	Non-speaking alternative activity
High streak, low accuracy	Slower pace, more revision
Multiple failures on same item	Insert into revision queue with scaffolded support

Personalisation Rules

- Recommendations are **transparent** — learner sees “Why this?” explanation.
- Learner can **override** recommendations.
- Personalisation does not create a filter bubble — new content is always introduced.

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33. AI Requirements

AI Feature	User Problem	Input	Output	Model Options	Non-AI Fallback	Priority
Lesson recommendations	“What should I learn next?”	Progress data, mastery scores	Next lesson/activity	Rule-based algorithm (no AI needed)	Static sequence	P0
Conversational practice	“I want to practice speaking”	Text/audio input	Contextual response	LLM API (OpenAI, Anthropic) with constrained prompts	Prerecorded dialogue	P1
Contextual hints	“I don’t understand this word”	Word, context	Definition, example	Rule-based dictionary + LLM for examples	Static dictionary	P1
Adaptive revision	“What should I review?”	Revision queue, mastery	Prioritized items	Spaced repetition algorithm	Static review list	P0
Pronunciation feedback	“How was my pronunciation?”	Audio recording	Feedback	Speech-to-text + comparison	Self-assessment	P2
Literary explanation	“Explain this Thirukkural”	Passage, language	Explanation, sources	LLM + RAG with verified sources	Curated explanations	P2

AI Safety Rules

- **No unconstrained LLM for foundational content** (letters, spelling, beginner vocabulary).
- AI responses for children are **filtered and constrained** with system prompts.
- **All AI interactions logged** for safety review (with parental consent).
- AI **never requests personal information** from the learner.
- AI **never pretends to be human** — clearly labeled as “Tamil helper.”

---

34. Audio and Speech Architecture

Audio Types

Type	Source	Format	Storage

Letter pronunciation	Prerecorded, reviewed	MP3, 128kbps	Supabase Storage
Word pronunciation	Prerecorded, reviewed	MP3	Supabase Storage
Sentence audio	Prerecorded, reviewed	MP3	Supabase Storage
Story narration	Prerecorded, reviewed	MP3	Supabase Storage
TTS fallback	Browser/cloud TTS	Streaming	Not stored

Audio Strategy

- **P0:** All core lesson audio is **prerecorded and reviewed** by a Tamil speaker.
- **P0:** Audio playback uses HTML5 Audio element with play/pause/repeat/speed controls.
- **P1:** Cloud TTS for dynamic content (e.g., AI-generated examples).
- **P2:** Speech recognition for pronunciation assessment (if validated API available).

Audio Failure Handling

Failure	User Message	Recovery
Audio fails to load	“Audio unavailable. Tap to retry.”	Retry button; text fallback
Browser blocks autoplay	“Tap to enable audio”	Manual play button
Microphone denied	“Microphone access needed for speaking practice”	Settings link; skip option

---

35. Global Localisation

Architecture

Use **structured translation dictionaries** with stable keys. No hardcoded UI strings. Recommended library: `react-i18next` or `react-intl`.

Translation File Structure

```
/locales
  /en
    common.json
    onboarding.json
    dashboard.json
    lessons.json
    errors.json
  /ta
    common.json
    onboarding.json
    ...
```

Localisation Coverage

UI Element	English	Tamil	Notes
Navigation labels	✓	✓	All labels
Buttons	✓	✓	All CTAs
Form labels	✓	✓	All inputs
Validation errors	✓	✓	All error messages
Notifications	✓	✓	In-app and push
Dashboard labels	✓	✓	All metrics
Lesson instructions	✓	✓	Per lesson
Accessibility labels	✓	✓	aria-label, alt text

Learning Content vs. UI

The **interface language** and **learning language** are independent. A learner can use the English interface while learning Tamil, or the Tamil interface while learning Tamil. The interface language never changes the target language.

---

36. UX Principles

Principle	Description	Implementation
Clarity	No ambiguity in instructions	Plain language; max 8 words per instruction
Consistency	Same action = same result	Design tokens for colors, spacing, typography
Feedback	Every action gets response	Visual confirmation within 100ms
Forgiveness	Mistakes are recoverable	Undo, retry, skip options
Progressive disclosure	Show only what's needed	Tanglish reduction; optional hints
Accessibility	Usable by all	WCAG 2.1 AA; keyboard navigation
Performance	Fast on low-end devices	<3s initial load; <1s interaction
Child safety	No harmful content	Content review; moderation

---

37. Visual Identity

Design Direction

The visual identity draws from **Tamil literary heritage** — palm-leaf manuscripts, temple architecture, and kolam patterns — interpreted through a modern educational design lens.

Proposed Palette

Token	Color	Hex	Usage
`--color-paper`	Warm Ivory	#FDF8F0	Background
`--color-ink`	Deep Charcoal	#1A1A2E	Primary text
`--color-teal`	Heritage Teal	#0F4C5C	Primary action
`--color-maroon`	Temple Maroon	#7B2D26	Accent, errors
`--color-gold`	Turmeric Gold	#E8B84B	Rewards, achievements
`--color-leaf`	Palm Green	#4A7C59	Success, correct
`--color-sand`	Soft Sand	#E8DCC8	Cards, surfaces
`--color-sky`	Morning Sky	#8ECAE6	Info, secondary

Typography

Usage	Font	Weight	Size
Headings	Noto Serif Tamil / Lora	600–700	24–48px
Body	Inter / Noto Sans	400–500	16–18px
Tamil script	Noto Sans Tamil	400–700	18–32px
Tanglish	Inter	400	14–16px
UI labels	Inter	500	14px

---

38. Design Tokens

```
{
  "spacing": { "xs": "4px", "sm": "8px", "md": "16px", "lg": "24px", "xl": "32px", "2xl": "48px" },
  "radius": { "sm": "4px", "md": "8px", "lg": "16px", "full": "9999px" },
  "shadow": {
    "sm": "0 1px 2px rgba(0,0,0,0.05)",
    "md": "0 4px 6px rgba(0,0,0,0.07)",
    "lg": "0 10px 15px rgba(0,0,0,0.1)"
  },
  "transition": { "fast": "150ms ease", "base": "250ms ease", "slow": "400ms ease" },
  "fontSize": { "xs": "12px", "sm": "14px", "base": "16px", "lg": "18px", "xl": "24px", "2xl": "32px" }
}
```

---

39. Typography

Font Stack

```
        --font-heading: 'Noto Serif Tamil', 'Lora', Georgia, serif;
        --font-body: 'Inter', 'Noto Sans', system-ui, sans-serif;
        --font-tamil: 'Noto Sans Tamil', 'Latha', sans-serif;
        --font-mono: 'JetBrains Mono', monospace;
```

Tamil Rendering Rules

- Use `font-feature-settings: "liga" 1, "kern" 1;` for proper ligatures.
- Set `line-height: 1.8` for Tamil text to accommodate vowel signs above and below.
- Test with `lang="ta"` attribute on Tamil text containers.
- Avoid `text-transform` on Tamil text.

---

40. Motion Design

Animation Principles

Principle	Description	Duration
Purposeful	Every animation explains a concept	—
Fast	No animation exceeds 400ms	150–400ms
Subtle	Movement supports, not distracts	2–8px displacement
Accessible	Respects `prefers-reduced-motion`	Instant fallback

Animation Types

Type	Usage	Duration	Easing
Fade	Screen transitions	200ms	ease-out
Slide	Modal entrance	250ms	ease-out
Scale	Button press	150ms	ease-in-out
Letter formation	Tamil letter animation	400ms	ease-in-out
Celebration	Correct answer	300ms	spring
Progress	Progress bar fill	400ms	ease-out

Reduced Motion

When `prefers-reduced-motion: reduce` is detected:

- Replace animated transitions with instant state changes.
- Disable letter formation animations (show static letters).
- Disable celebration particles.
- Keep progress bar updates but remove animation.

---

41. 3D Visual Strategy

Where 3D Helps

Context	3D Approach	Justification
Letter formation	3D letter objects that rotate/assemble	Spatial understanding of letter structure
Word building	3D tiles snapping together	Physical metaphor for construction
Level progression	3D path with milestones	Engaging progress visualization
Celebration	Lightweight particle burst	Reward feedback

Where 2D Is Better

Context	2D Approach	Justification
Reading passages	Flat text layout	Reading comfort
Quiz interfaces	Flat cards	Clarity, speed
Dashboard	Flat charts/graphs	Data readability
Settings	Flat forms	Convention

Implementation

- **P0:** CSS transforms and SVG animations for letter tiles and transitions.
- **P1:** Three.js for letter formation objects (lazy-loaded).
- **Fallback:** Static images when WebGL not supported or reduced motion enabled.

---

42. Accessibility

WCAG 2.1 AA Compliance Targets

Criterion	Target	Implementation
Color contrast	4.5:1 normal text, 3:1 large text	Design token validation
Keyboard navigation	All interactive elements	Focus indicators, tab order
Screen reader	All content accessible	Semantic HTML, ARIA labels
Touch targets	44×44px minimum	Button sizing
Text resizing	Up to 200% without loss	Responsive typography
Motion	Reduced motion support	`prefers-reduced-motion`
Audio	Captions for all audio	Text alternatives
Language	`lang` attribute set	Per-element language marking

Tamil-Specific Accessibility

- Screen readers must handle Tamil Unicode correctly (test with NVDA, VoiceOver).
- Provide text alternatives for all Tamil audio content.
- Allow slower audio playback (0.5x) for processing difficulties.
- Ensure Tamil vowel signs are not clipped at line boundaries.

---

43. Responsive Design

Breakpoints

Breakpoint	Width	Layout
Mobile	< 640px	Single column, bottom navigation
Tablet	640–1024px	Two column, side navigation
Desktop	> 1024px	Three column, top navigation

Responsive Behavior

Element	Mobile	Tablet	Desktop
Lesson player	Full screen	Centered card	Split view
Dashboard	Stacked cards	2-column grid	3-column grid
Navigation	Bottom tabs	Side drawer	Top bar
Audio controls	Sticky bottom	Inline	Inline

---

44. Information Architecture

```
TamilPalam
├── Landing
│   ├── Hero
│   ├── How it works
│   ├── Language selector
│   └── CTA (Sign up / Demo)
├── Auth
│   ├── Sign up
│   ├── Sign in
│   └── Parental consent
├── Onboarding
│   ├── Welcome
│   ├── Assessment
│   └── Level recommendation
├── Dashboard
│   ├── Daily goal
│   ├── Continue learning
│   ├── Progress summary
│   └── Achievements
├── Learn
│   ├── Level map
│   ├── Lesson player
│   │   ├── Letter learning
│   │   ├── Vocabulary
│   │   ├── Sentence building
│   │   ├── Listening
│   │   ├── Speaking
│   │   └── Reading
│   └── Quiz results
├── Progress
│   ├── Skill mastery
│   ├── Streak calendar
│   └── Badge collection
├── Profile
│   ├── Settings
│   ├── Language preference
│   └── Account management
└── Parent Dashboard (P1)
    ├── Child progress
    └── Weekly report
```

---

45. Navigation Model

Navigation Type	Mobile	Desktop
Primary	Bottom tab bar (Home, Learn, Progress, Profile)	Top nav bar
Secondary	Hamburger menu	Sidebar
In-lesson	Progress bar + exit button	Same
Modal	Bottom sheet	Centered dialog

---

46. Full Screen Inventory

#	Screen	Purpose	Priority
1	Landing	Introduce product, capture interest	P0
2	Sign up / Sign in	Authenticate user	P0
3	Language selector	Choose English/Tamil interface	P0
4	Onboarding	Explain platform, collect minimal info	P0
5	Initial assessment	Determine starting level	P0
6	Dashboard	Overview of progress and next actions	P0
7	Level selection	Choose learning level	P0
8	Lesson player	Deliver learning content	P0
9	Tamil letter screen	Learn individual letters	P0

10	Vocabulary screen	Learn words with audio	P0
11	Sentence-building screen	Construct sentences	P0
12	Listening activity	Audio comprehension	P0
13	Speaking activity	Pronunciation practice	P0
14	Reading activity	Reading comprehension	P0
15	Quiz results	Show assessment results	P0
16	Progress & achievements	Show badges, mastery, streak	P0
17	Profile & settings	Manage account	P0
18	Parent dashboard	View child progress	P1
19	Empty/loading/error/offline	Handle edge states	P0
20	Research assistant	Literary explanation	P2

---

47. Wireframes (Text-Based)

Screen 6: Dashboard

```
+-----------------------------------------+
| [Logo] TamilPalam    [EN/TA] [Profile]  |
+-----------------------------------------+
|                                         |
|  Welcome back, Aarav!                   |
|  🔥 5-day streak                        |
|                                         |
|  +-----------------------------------+  |
|  | Daily Goal: 1 lesson              |  |
|  | [========>          ] 1/1         |  |
|  | [Continue Learning]               |  |
|  +-----------------------------------+  |
|                                         |
|  +-----------------------------------+  |
|  | Current Level: Level 2            |  |
|  | Progress: 60%                     |  |
|  | [==========>       ]              |  |
|  | [View Level Map]                  |  |
|  +-----------------------------------+  |
|                                         |
|  Recent Achievements                    |
|  [Badge: 10 Words] [Badge: 3-day streak]|
|                                         |
+-----------------------------------------+
```

Screen 8: Lesson Player

```
+-----------------------------------------+
| [← Exit]   Lesson 3 of 8    [========>] |
+-----------------------------------------+
|                                         |
|  Tamil Letter                           |
|                                         |
|         அ                               |
|                                         |
|  [🔊 Play Audio]                        |
|                                         |
|  Tanglish: a                            |
|                                         |
|  +-----------------------------------+  |
|  | Which letter makes the sound "a"?  |  |
|  |                                   |  |
|  |  [அ]  [ஆ]  [இ]  [ஈ]             |  |
|  +-----------------------------------+  |
|                                         |
|  [Need help?]                           |
|                                         |
+-----------------------------------------+
```

---

48. Detailed Screen Specifications

Screen 1: Landing Page

Attribute	Specification
Purpose	Introduce TamilPalam, communicate value, drive sign-up
Target User	Parents, older learners
Layout	Hero section, 3 feature cards, testimonial, CTA
Components	Logo, tagline, language toggle, “How it works” section, Sign up button, Demo button
Actions	Toggle language, scroll to features, click CTA
States	Default; language selected
Navigation	Sign up → Auth; Demo → Guest dashboard
Responsive	Mobile: stacked; Desktop: 2-column
Accessibility	All images have alt text; keyboard navigable
Data	None (static)
Acceptance	Page loads <2s; language toggle works; CTA navigates correctly

Screen 8: Lesson Player

Attribute	Specification
Purpose	Deliver interactive learning content
Target User	All learners
Layout	Progress bar (top), content area (center), controls (bottom)
Components	Progress indicator, letter/word display, audio button, Tanglish toggle, exercise area, feedback area, continue button
Actions	Play audio, select answer, request hint, continue, exit
States	Loading, presenting, awaiting response, correct feedback, incorrect feedback, complete
Navigation	Exit → Dashboard; Complete → Quiz results
Responsive	Full width on mobile; centered 800px on desktop
Accessibility	Screen reader announces content; keyboard selectable; focus management
Data	Lesson items, user progress, audio assets
Acceptance	Lesson completes within 10 min; progress saved after each item; resume works

---

49. User Journeys

Main User Journey

```
flowchart TD
    A[Visit Landing Page] --> B{Choose Interface Language}
    B -->|English| C[Read English Content]
    B -->|Tamil| D[Read Tamil Content]
    C --> E[Click Sign Up]
    D --> E
    E --> F[Create Account or Demo]
    F --> G[Onboarding: How it Works]
    G --> H[Initial Assessment]
    H --> I{Assessment Complete?}
    I -->|Yes| J[Receive Level Recommendation]
    I -->|Skip| K[Start Level 0]
    J --> L[Dashboard]
    K --> L
    L --> M[Open Lesson]
    M --> N[Complete Lesson Items]
    N --> O{Item Correct?}
    O -->|Yes| P[Positive Feedback + XP]
    O -->|No| Q[Hint or Correct Answer]
    P --> R[Continue to Next Item]
    Q --> R
    R --> S{Lesson Complete?}
    S -->|No| N
    S -->|Yes| T[Update Progress]
    T --> U[Update Dashboard]
    U --> V{Streak Maintained?}
    V -->|Yes| W[Streak Increment]
    V -->|No| X[Streak Reset with Grace]
    W --> Y[Recommend Next Lesson]
    X --> Y
    Y --> Z[Return to Dashboard or Continue]
```

Authentication Flow

```
flowchart TD
    A[User Clicks Sign Up] --> B{Has Account?}
    B -->|No| C[Enter Email + Password]
    B -->|Yes| D[Enter Email + Password]
    C --> E[Validate Input]
    D --> E
    E --> F{Valid?}
    F -->|No| G[Show Error Message]
    F -->|Yes| H{Email Confirmation Required?}
    H -->|Yes| I[Send Confirmation Email]
    H -->|No| J[Create Session]
    I --> K[User Confirms Email]
    K --> J
    J --> L{Under 13?}
    L -->|Yes| M[Request Parental Consent]
    L -->|No| N[Access Dashboard]
    M --> O{Consent Given?}
    O -->|Yes| N
    O -->|No| P[Limited Demo Mode]
```

Parent Progress Review Flow

```
flowchart TD
    A[Parent Logs In] --> B[View Linked Children]
    B --> C[Select Child]
    C --> D[View Progress Dashboard]
    D --> E{What to View?}
    E -->|Overview| F[Weekly Summary]
    E -->|Details| G[Skill Mastery Breakdown]
    E -->|Activity| H[Lesson History]
    F --> I[Email Report Option]
    G --> J[Export CSV]
    H --> K[View Lesson Details]
    I --> L[Done]
    J --> L
    K --> L
```

---

50. Workflow Diagrams

Lesson Interaction Flow

```
sequenceDiagram
    participant Learner
    participant LessonPlayer
    participant AudioService
    participant ProgressService
    participant Database

    Learner->>LessonPlayer: Open lesson
    LessonPlayer->>Database: Fetch lesson items
    Database-->>LessonPlayer: Return items
    LessonPlayer->>Learner: Display item 1
    Learner->>LessonPlayer: Play audio
    LessonPlayer->>AudioService: Request audio
    AudioService-->>LessonPlayer: Stream audio
    LessonPlayer->>Learner: Play audio
    Learner->>LessonPlayer: Submit answer
    LessonPlayer->>LessonPlayer: Evaluate answer
    alt Correct
        LessonPlayer->>Learner: Show correct feedback
        LessonPlayer->>ProgressService: Record correct
    else Incorrect
        LessonPlayer->>Learner: Show hint
        Learner->>LessonPlayer: Retry
        LessonPlayer->>ProgressService: Record attempt
    end
    LessonPlayer->>Learner: Show next item
    Note over Learner,Database: Repeat for all items
    LessonPlayer->>Database: Save lesson completion
    Database-->>LessonPlayer: Confirm
    LessonPlayer->>Learner: Show lesson complete
```

---

51. System Architecture

Overview



Architecture Decisions

Decision	Choice	Rationale	Alternatives Considered
Frontend framework	React + TypeScript	Component model, ecosystem, AI tool support	Vue (smaller ecosystem), Svelte (less mature tooling)
Build tool	Vite	Fast HMR, simple config, TypeScript support	Webpack (slower, complex config)
CSS framework	Tailwind CSS	Utility-first, design token friendly, small bundle	CSS Modules (more verbose), styled-components (runtime cost)
Backend	Supabase	Auth + DB + Storage + RLS in one; PostgreSQL; generous free tier	Firebase (NoSQL, less relational), custom Express (more work)
Database	PostgreSQL (via Supabase)	Relational integrity, RLS, JSON support, mature	MongoDB (no RLS), SQLite (not serverless)
Auth	Supabase Auth	Integrated with RLS, OAuth support, email/password	Auth0 (expensive), custom JWT (more work)
File storage	Supabase Storage	Integrated with auth, CDN, simple API	S3 (complex setup), Cloudinary (extra service)
Hosting	Vercel / Netlify	Free tier, Git integration, CDN	AWS Amplify (complex), self-hosted (maintenance)

Rejected architecture components:

- **FastAPI separate backend:** Not justified. Supabase client + RLS handles all MVP needs. Edge functions for protected operations only.
- **Vector database:** Not needed. No semantic search in MVP.
- **Graph database:** Not needed. Relational model sufficient.
- **Multiple AI frameworks:** One LLM provider API maximum for P1 features.

---

52. Technology Decisions

Category	Technology	Version	Justification
Language	TypeScript	5.x	Type safety, AI tool compatibility
Framework	React	18.x	Component model, hooks, ecosystem
Build	Vite	5.x	Speed, simplicity, TypeScript
Styling	Tailwind CSS	3.x	Utility-first, design tokens
i18n	react-i18next	Latest	Structured dictionaries, interpolation
Routing	React Router	6.x	Declarative, nested routes
State	Zustand / React Query	Latest	Lightweight; server state caching
Forms	React Hook Form + Zod	Latest	Performance, validation
Audio	HTML5 Audio + Howler.js	Latest	Cross-browser, sprite support
Animation	Framer Motion	Latest	Declarative, reduced-motion support
Charts	Recharts	Latest	React-native, responsive
Testing	Vitest + Playwright	Latest	Unit + E2E
Linting	ESLint + Prettier	Latest	Code quality
Backend	Supabase	Latest	Auth, DB, Storage, RLS
Database	PostgreSQL	15+	Via Supabase
Deployment	Vercel	—	Zero-config, CDN
Monitoring	Sentry	—	Error tracking (free tier)

---

53. Frontend Architecture

Component Structure

```
src/
├── app/                    # Route definitions
│   ├── routes.tsx
│   └── App.tsx
├── components/
│   ├── ui/                 # Reusable UI primitives
│   │   ├── Button.tsx
│   │   ├── Card.tsx
│   │   ├── Input.tsx
│   │   ├── Modal.tsx
│   │   ├── ProgressBar.tsx
│   │   └── AudioPlayer.tsx
│   ├── layout/             # Layout components
│   │   ├── AppShell.tsx
│   │   ├── BottomNav.tsx
│   │   └── TopNav.tsx
│   └── features/           # Feature-specific components
│       ├── auth/
│       ├── dashboard/
│       ├── lesson/
│       ├── progress/
│       └── parent/
├── hooks/                  # Custom React hooks
│   ├── useAuth.ts
│   ├── useLesson.ts
│   ├── useProgress.ts
│   └── useAudio.ts
├── lib/                    # Utilities and clients
│   ├── supabase.ts
│   ├── i18n.ts
│   └── audio.ts
├── stores/                 # Zustand stores
│   ├── authStore.ts
│   ├── lessonStore.ts
│   └── uiStore.ts
├── types/                  # TypeScript types
│   ├── database.ts
│   ├── lesson.ts
│   └── user.ts
├── locales/                # Translation files
│   ├── en/
│   └── ta/
└── styles/                 # Global styles
    └── globals.css
```

Key Patterns

- **Server state:** React Query for data fetching, caching, and mutation.
- **Client state:** Zustand for UI state (lesson progress, modals).
- **Auth:** Supabase Auth context provider wrapping app.
- **i18n:** react-i18next with lazy-loaded translation files.
- **Error boundaries:** Per-feature error boundaries with fallback UI.
- **Code splitting:** Route-based lazy loading.

---

54. Backend Architecture

Supabase Services Used

Service	Usage	Priority
Auth	Email/password, OAuth (Google), session management	P0
PostgreSQL	All relational data	P0
Row Level Security	Data isolation per user	P0
Storage	Audio files, images	P0
Edge Functions	Protected operations (AI calls, admin actions)	P1
Realtime	Live progress updates (optional)	P2

API Strategy

The MVP uses **Supabase client operations** directly from the frontend for:

- User profile CRUD (with RLS)
- Lesson data fetching
- Progress recording
- Achievement retrieval

Edge Functions used only for:

- AI provider API calls (P1) — keeps API keys server-side
- Admin operations (content management)
- Webhook handling (payment, if added later)

No custom REST API server is required for the MVP.

---

55. Database Architecture

Design Principles

- **Normalized relational schema** with foreign key constraints.
- **RLS on every table** with both `USING` and `WITH CHECK` policies.
- **UUID primary keys** for all user-facing tables.
- **Timestamps** (`created_at`, `updated_at`) on all tables.
- **Soft deletes** where data recovery is needed.
- **Indexes** on all foreign keys and frequent query columns.

---

56. ER Diagram

```
erDiagram
    profiles ||--o{ user_preferences : has
    profiles ||--o{ lesson_progress : tracks
    profiles ||--o{ quiz_attempts : makes
    profiles ||--o{ learner_skill_mastery : has
    profiles ||--o{ user_achievements : earns
    profiles ||--o{ daily_learning_activity : logs
    profiles ||--o{ parent_child_relationships : "parent of"
    learning_levels ||--o{ lessons : contains
    lessons ||--o{ lesson_items : contains
    lessons ||--o{ exercises : has
    exercises ||--o{ questions : contains
    questions ||--o{ answer_options : has
    tamil_letters ||--o{ pronunciation_assets : has
    vocabulary_items ||--o{ pronunciation_assets : has
    lesson_items ||--o{ vocabulary_items : references
    lesson_items ||--o{ tamil_letters : references
    achievements ||--o{ user_achievements : earned
```

---

57. Data Dictionary

profiles

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Profile ID
user_id	UUID	FK → auth.users, UNIQUE, NOT NULL	Supabase auth user
display_name	TEXT	NOT NULL	Learner display name
interface_language	TEXT	CHECK IN (‘en’,‘ta’), DEFAULT ‘en’	UI language
date_of_birth	DATE	NULL	For age-based features
is_child	BOOLEAN	DEFAULT false	Under-13 flag

parental_consent	BOOLEAN	DEFAULT false	Consent status
created_at	TIMESTAMPTZ	DEFAULT now()	
updated_at	TIMESTAMPTZ	DEFAULT now()	

user_preferences

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles, UNIQUE	
daily_goal	INTEGER	DEFAULT 1, CHECK 1–3	Lessons per day
audio_speed	NUMERIC	DEFAULT 1.0	Playback speed
reduced_motion	BOOLEAN	DEFAULT false	Accessibility
tanglish_enabled	BOOLEAN	DEFAULT true	Manual override
notifications_enabled	BOOLEAN	DEFAULT true	

learning_levels

Column	Type	Constraints	Description
id	UUID	PK	
level_number	INTEGER	UNIQUE, 0–7	Level number
title_en	TEXT	NOT NULL	English title
title_ta	TEXT	NOT NULL	Tamil title
description_en	TEXT		
description_ta	TEXT		
prerequisite_level	UUID	FK → learning_levels	

lessons

Column	Type	Constraints	Description
id	UUID	PK	
level_id	UUID	FK → learning_levels	
title_en	TEXT	NOT NULL	
title_ta	TEXT	NOT NULL	
sequence_order	INTEGER	NOT NULL	
lesson_type	TEXT	CHECK IN ('letters','vocabulary','sentences','listening','speaking','reading','story')	
estimated_minutes	INTEGER	DEFAULT 10	
tanglish_stage	TEXT	CHECK IN ('A','B','C','D','E')	Required stage

lesson_items

Column	Type	Constraints	Description
id	UUID	PK	
lesson_id	UUID	FK → lessons	
item_type	TEXT	CHECK IN ('letter','word','sentence','passage')	
content_ta	TEXT	NOT NULL	Tamil script
content_tanglish	TEXT		Transliteration
content_en	TEXT		English meaning
audio_url	TEXT		Storage path
image_url	TEXT		Storage path

sequence_order	INTEGER		

tamil_letters

Column	Type	Constraints	Description
id	UUID	PK	
letter	TEXT	UNIQUE, NOT NULL	Tamil character
unicode_codepoint	TEXT		e.g., U+0B85
letter_type	TEXT	CHECK IN ('vowel','consonant','vowel_sign','virama')	
tanglish_primary	TEXT		
tanglish_alternatives	TEXT[]		
ipa	TEXT		IPA notation
example_word_ta	TEXT		
example_word_en	TEXT		

exercises

Column	Type	Constraints	Description
id	UUID	PK	
lesson_id	UUID	FK → lessons	
exercise_type	TEXT	CHECK IN ('mcq','matching','fill_blank','ordering','speaking','listening')	
prompt_en	TEXT		
prompt_ta	TEXT		
sequence_order	INTEGER		

questions

Column	Type	Constraints	Description
id	UUID	PK	
exercise_id	UUID	FK → exercises	
question_text	TEXT	NOT NULL	
correct_answer	TEXT	NOT NULL	
explanation	TEXT		Shown on incorrect

answer_options

Column	Type	Constraints	Description
id	UUID	PK	
question_id	UUID	FK → questions	
option_text	TEXT	NOT NULL	
is_correct	BOOLEAN	DEFAULT false	
sequence_order	INTEGER		

lesson_progress

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
lesson_id	UUID	FK → lessons	
status	TEXT	CHECK IN ('not_started','in_progress','completed')	

score	NUMERIC	0–100	
attempts	INTEGER	DEFAULT 0	
last_item_index	INTEGER		For resume
completed_at	TIMESTAMPTZ		
UNIQUE(profile_id, lesson_id)			

quiz_attempts

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
lesson_id	UUID	FK → lessons	
score	NUMERIC		
correct_count	INTEGER		
total_count	INTEGER		
time_spent_seconds	INTEGER		
created_at	TIMESTAMPTZ	DEFAULT now()	

learner_skill_mastery

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
skill_type	TEXT	CHECK IN ('letter_recognition','word_reading','sentence_comprehension','listening','speaking')	
skill_item_id	TEXT		ID of letter/word
mastery_score	NUMERIC	0–1	
attempts	INTEGER		
correct	INTEGER		
last_attempted_at	TIMESTAMPTZ		
UNIQUE(profile_id, skill_type, skill_item_id)			

revision_queue

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
item_type	TEXT		
item_id	TEXT		
next_review_at	TIMESTAMPTZ		
interval_days	INTEGER	DEFAULT 1	
ease_factor	NUMERIC	DEFAULT 2.5	
consecutive_correct	INTEGER	DEFAULT 0	

achievements

Column	Type	Constraints	Description
id	UUID	PK	
name_en	TEXT	NOT NULL	

name_ta	TEXT	NOT NULL	
description_en	TEXT		
description_ta	TEXT		
icon	TEXT		Icon identifier
criteria_type	TEXT		e.g., ‘streak_days’,‘words_learned’
criteria_value	INTEGER		

user_achievements

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
achievement_id	UUID	FK → achievements	
earned_at	TIMESTAMPTZ	DEFAULT now()	
UNIQUE(profile_id, achievement_id)			

daily_learning_activity

Column	Type	Constraints	Description
id	UUID	PK	
profile_id	UUID	FK → profiles	
activity_date	DATE	NOT NULL	
lessons_completed	INTEGER	DEFAULT 0	
minutes_spent	INTEGER	DEFAULT 0	
xp_earned	INTEGER	DEFAULT 0	
UNIQUE(profile_id, activity_date)			

parent_child_relationships

Column	Type	Constraints	Description
id	UUID	PK	
parent_profile_id	UUID	FK → profiles	
child_profile_id	UUID	FK → profiles	
relationship_type	TEXT		‘parent’,‘guardian’
verified	BOOLEAN	DEFAULT false	
created_at	TIMESTAMPTZ		
UNIQUE(parent_profile_id, child_profile_id)			

---

58. Data Lifecycle

Data Type	Retention	Deletion Method	Trigger
Profile	Until account deletion	Cascade delete	User/parent request
Lesson progress	Until account deletion	Cascade delete	Account deletion
Quiz attempts	Until account deletion	Cascade delete	Account deletion
Audio recordings (speaking)	24 hours	Automatic	Timer
AI interaction logs	90 days	Automatic	Timer
Analytics events	12 months	Anonymized	Timer
Parent relationship	Until unlinked	Manual	Parent/child request

---

59. API Specification

Supabase Client Operations (Primary)

Operation	Method	Table	Auth	Description
Get profile	SELECT	profiles	Required	Fetch own profile
Update profile	UPDATE	profiles	Required	Update own profile
Get lessons	SELECT	lessons	Required	Fetch lessons for level
Get lesson items	SELECT	lesson_items	Required	Fetch items for lesson
Record progress	INSERT/UPDATE	lesson_progress	Required	Save progress
Get achievements	SELECT	user_achievements	Required	Fetch earned badges
Get daily activity	SELECT	daily_learning_activity	Required	Fetch activity log
Get revision queue	SELECT	revision_queue	Required	Fetch due items

Edge Functions (Protected)

Function	Method	Auth	Description	Priority
`ai-hint`	POST	Required + rate limit	AI hint generation	P1
`ai-conversation`	POST	Required + rate limit	Conversational practice	P1
`admin-content`	POST	Admin only	Content management	P1
`parent-report`	POST	Parent only	Generate progress report	P1

Security: All Edge Functions validate JWT, check RLS-equivalent permissions server-side, and never expose API keys to the client.

---

60. Authentication

Supabase Auth Configuration

Setting	Value	Rationale
Email/password	Enabled	Primary auth
Google OAuth	Enabled	Convenience
Apple OAuth	Enabled	iOS users
Email confirmation	Required for production; disabled for demo	Security vs. demo speed
Password minimum	8 characters	Security
Session expiry	7 days (refresh token)	Balance security and UX
MFA	Optional (P1)	Child accounts often managed by parents

Child Account Flow

```
Child signs up → Age entered → If <13:
  → Parent email requested
  → Consent email sent to parent
  → Child can access demo mode only
  → Parent clicks consent link
  → Child account activated with full access
  → Parent can view progress
```

---

61. Authorisation

Role-Based Access Control

Role	Permissions	Assignment
anon	Read public content (landing)	Unauthenticated
authenticated	Full own-data CRUD	Any logged-in user
parent	View linked child's progress	Parent relationship
teacher	View assigned students (P2)	Teacher relationship
admin	Content management, user management	Internal only

RLS Policy Examples

```
-- Profiles: Users can read/update their own
CREATE POLICY "Users can view own profile"
ON profiles FOR SELECT
TO authenticated
USING (auth.uid() = user_id);

CREATE POLICY "Users can update own profile"
ON profiles FOR UPDATE
TO authenticated
USING (auth.uid() = user_id)
WITH CHECK (auth.uid() = user_id);

-- Lesson progress: Users can CRUD own progress
CREATE POLICY "Users can view own progress"
ON lesson_progress FOR SELECT
TO authenticated
USING (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));

CREATE POLICY "Users can insert own progress"
ON lesson_progress FOR INSERT
TO authenticated
WITH CHECK (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));

-- Parent access to child progress
CREATE POLICY "Parents can view child progress"
ON lesson_progress FOR SELECT
TO authenticated
USING (profile_id IN (
  SELECT child_profile_id FROM parent_child_relationships
  WHERE parent_profile_id IN (
    SELECT id FROM profiles WHERE user_id = auth.uid()
  ) AND verified = true
));
```

---

62. Security

Security Layer	Implementation
Authentication	Supabase Auth with JWT, email confirmation, OAuth
Authorisation	Row Level Security on all tables; role-based policies
API security	Edge Functions validate JWT; no service_role key in client
Data encryption	TLS in transit; Supabase encrypts at rest
Input validation	Zod schemas on client; Supabase constraints on server
Rate limiting	Supabase auth rate limits; Edge Function custom limits
CORS	Restricted to application domain
Secrets	Environment variables; never committed; Edge Function env
Dependency scanning	npm audit; Dependabot
Penetration testing	Basic OWASP checklist before launch

Critical rule: The `SUPABASE_SERVICE_ROLE_KEY` is **never** exposed to the client. All privileged operations go through Edge Functions.

---

63. Privacy

Data Minimisation

Data Type	Collected?	Purpose	Retention
Display name	Yes	Personalization	Until deletion
Email	Yes	Auth, communication	Until deletion
Date of birth	Optional	Age-appropriate content	Until deletion
Learning progress	Yes	Core functionality	Until deletion
Audio recordings	Temporary	Speaking practice	24 hours
Device info	Minimal	Error monitoring	90 days
Location	No	—	—
Photos	No	—	—

Privacy Policy Requirements

- Clear, plain-language privacy policy accessible from every screen.
- Separate children's privacy section.
- Parental rights: access, correction, deletion.
- Data processing purposes listed.
- Third-party services disclosed (Supabase, AI provider if used).
- Contact for privacy inquiries.

---

64. Child Safety

COPPA 2.0 Compliance (Effective June 2025; Full Compliance April 2026)

Requirement	Implementation
Verifiable parental consent	Text-plus method (SMS + secondary confirmation) or email verification
Data minimisation	Collect only display name and progress data
No behavioural advertising	Zero ad tracking for children
Data retention limits	Delete recordings after 24 hours; anonymize analytics after 12 months
Parental access rights	Parent dashboard with full progress visibility
Parental deletion rights	Account deletion cascades all data
Third-party SDK compliance	Audit all embedded SDKs; no non-essential third parties
Voice data protection	Recordings not used for training; deleted after use
No monetisation of child data	Strict policy; no ads, no data sales

GDPR-K Compliance (EU/UK)

Requirement	Implementation
Age-appropriate consent	Consent age varies by country (13–16); geo-detect and apply strictest
Verifiable consent	Parent must confirm via email + secondary step
Right to erasure	Full account deletion within 30 days
Data portability	Export progress data as JSON
Transparency	Child-friendly privacy summary in addition to full policy

Safe Defaults

- **Public profiles:** Disabled by default. No social features in MVP.
- **Leaderboards:** Not implemented (child safety).

- **Notifications:** Opt-in only; parent controls for child accounts.
- **AI interactions:** Logged and reviewable; no personal information requested.

---

65. Content Management Strategy

Content Creation Workflow

```
Curriculum Designer → Content Draft → Tamil Reviewer → Audio Recording → QA → Publication
```

Role	Responsibility	Tools
Curriculum Designer	Create lessons, exercises, vocabulary lists	Google Sheets / CMS
Tamil Reviewer	Verify spelling, grammar, transliteration	Review checklist
Audio Recorder	Record pronunciation audio	Audacity / Studio
QA	Test rendering, playback, exercises	Test cases

Content Types

Type	Source	Review Required	Frequency
Tamil letters	Unicode standard	Tamil speaker	One-time
Vocabulary	Curriculum designer	Tamil speaker	Per batch
Sentences	Curriculum designer	Tamil speaker	Per batch
Stories	Tamil literature / original	Tamil speaker + cultural reviewer	Per story
Audio	Native speaker recording	Audio QA	Per recording

---

66. Content Verification

Verification Checklist

- [ ] Tamil spelling verified by native speaker
- [ ] Letter-sound association verified
- [ ] Transliteration reviewed for consistency
- [ ] English meaning accurate and natural
- [ ] Example sentences grammatically correct
- [ ] Cultural references appropriate and respectful
- [ ] Audio matches text exactly
- [ ] Audio pronunciation natural and clear
- [ ] No regional dialect bias (or dialect explicitly noted)
- [ ] Content age-appropriate

Error Reporting

- In-app “Report an issue” button on every lesson item.
- Reports routed to content team with item ID and screenshot.
- Correction history tracked in `content_versions` table (P1).
- Verified corrections pushed to production within 48 hours.

---

67. Analytics

Events Tracked

Event	Properties	Purpose	Privacy
`app_open`	language, session_id	Usage	Anonymized
`lesson_start`	lesson_id, level	Engagement	User-linked

`lesson_complete`	lesson_id, score, time	Learning	User-linked
`item_answer`	item_id, correct, attempts	Skill mastery	User-linked
`audio_play`	item_id, type	Audio engagement	User-linked
`speaking_record`	item_id, duration	Speaking practice	Not stored
`streak_update`	streak_days	Motivation	User-linked
`achievement_earned`	achievement_id	Motivation	User-linked
`error_occurred`	error_type, screen	Reliability	Anonymized

Analytics Privacy

- No third-party analytics SDKs in MVP.
- All events stored in Supabase `analytics_events` table (P1).
- Parent can opt out of analytics for child accounts.
- IP addresses not stored.

---

68. Success Metrics

Metric	Formula	Data Source	Frequency	Target	Limitation
Onboarding completion	Users completing assessment / users starting	Event logs	Weekly	>70%	Doesn't measure learning
First lesson completion	Users completing 1 lesson / onboarded	Progress table	Weekly	>80%	—
Weekly active learners	Unique users with ≥1 session / total	Activity table	Weekly	>40%	—
Lessons per learner	Total lessons / active learners	Progress table	Weekly	≥3	—
Session duration	Avg time per session	Activity table	Weekly	8–15 min	Time ≠ learning
Return rate (D7)	Users active on day 7 / day 1	Activity table	Weekly	>30%	—
Quiz accuracy	Correct / total answers	Quiz attempts	Weekly	>75%	—
Vocabulary retention	Words recalled after 7 days / words learned	Skill mastery	Weekly	>70%	—
Reading accuracy	Correctly read words / total	Reading exercises	Weekly	>80%	Self-reported
Tanglish reduction	Learners advancing stages / total	Mastery data	Monthly	>60% reach C	—
Speaking participation	Users recording / users in speaking lessons	Event logs	Weekly	>50%	—
Parent engagement	Parents viewing dashboard / parents linked	Parent table	Monthly	>40%	P1

Critical disclaimer: Time spent in the application is **not** equivalent to learning. Quiz accuracy and retention metrics are the primary indicators of learning progress.

---

69. Functional Requirements

FR-001 — Global Interface Language Switching

Description: When a user selects English or Tamil in the global language selector, every supported interface string must update to the selected language without resetting the user's lesson progress.

Priority: P0

User Story: As a learner, I want to switch the interface language so that I can use the app in my preferred language.

Preconditions: User is on any authenticated screen with language selector visible.

Trigger: User taps language selector and chooses English or Tamil.

Main Flow: 1. User taps language selector. 2. Dropdown/modal shows “English” and “தமிழ்” options. 3. User selects one. 4. All UI strings update immediately. 5. Selection persists in `user_preferences`.

Alternative Flows: - User cancels selection → no change. - Selected language is already active → no change.

Error Cases: - Missing translation key → fallback to English; log warning. - Network offline → apply locally; sync when online.

Data Changes: `user_preferences.interface_language` updated.

Security: RLS ensures user can only update own preference.

Acceptance Criteria: - All navigation labels use selected language. - All buttons and validation messages use selected language. - Current lesson remains open after switching. - Language persists across sessions. - Tamil learning content unchanged by UI language. - Missing translations detected in development.

Dependencies: i18n library configured; translation files complete.

Test Scenarios: - Switch from EN to TA on dashboard → all labels change. - Switch mid-lesson → lesson content remains, UI changes. - Refresh page → language persists.

FR-002 — User Onboarding

Description: New users complete a guided onboarding flow that explains the platform and collects minimal information.

Priority: P0

User Story: As a new user, I want to understand how the platform works before starting.

Main Flow: 1. User lands on onboarding screen. 2. Screen 1: Welcome + how it works (3 slides). 3. Screen 2: Display name input. 4. Screen 3: Date of birth (optional) → age detection. 5. If under 13: Parental consent flow. 6. Screen 4: Initial assessment intro. 7. User completes or skips assessment.

Acceptance Criteria: - Onboarding completes in <2 minutes. - Only display name and optional DOB collected. - Under-13 users cannot proceed without parental consent. - Skip option available for assessment.

FR-003 — Initial Assessment

Description: Optional assessment determines starting level.

Priority: P0

Main Flow: 1. System presents 10 adaptive questions. 2. Questions range from letter recognition to sentence comprehension. 3. System calculates starting level (0–7). 4. Recommendation shown with option to override.

Acceptance Criteria: - Assessment completes in <5 minutes. - Recommendation is reasonable (not too high/low). - User can choose different level.

FR-004 — Lesson Playback

Description: Learner can play, pause, repeat, and adjust speed of audio.

Priority: P0

Main Flow: 1. Audio auto-plays on item display (if enabled). 2. User can tap play/pause. 3. User can tap repeat. 4. Speed control (0.5x, 0.75x, 1x) available.

Acceptance Criteria: - Audio loads <2 seconds. - Controls respond <200ms. - Speed change applies immediately.

FR-005 — Progress Persistence

Description: Lesson progress is saved after each item and survives navigation, refresh, and device change.

Priority: P0

Main Flow: 1. User answers item. 2. System saves progress to `lesson_progress`. 3. User exits lesson. 4. User returns → lesson resumes at last item.

Acceptance Criteria: - Progress saved within 1 second of item completion. - Resume works after browser refresh. - Progress syncs across devices (if logged in).

---

70. Non-Functional Requirements

ID	Requirement	Target	Validation Method
NFR-001	Initial load time	<3s on 3G	Lighthouse
NFR-002	Interaction response	<200ms	Performance API
NFR-003	Audio load	<2s	Network timing
NFR-004	Accessibility	WCAG 2.1 AA	axe-core audit
NFR-005	Uptime	99.5%	Monitoring
NFR-006	Data durability	99.99%	Supabase SLA
NFR-007	Mobile responsiveness	Works on 320px width	Device testing
NFR-008	Browser support	Chrome, Safari, Firefox, Edge (last 2 versions)	Cross-browser testing
NFR-009	Low-bandwidth	Functional on 2G	Throttled testing
NFR-010	Error recovery	No data loss on network failure	Chaos testing
NFR-011	Tamil rendering	Correct in all supported browsers	Visual regression
NFR-012	Security	No critical vulnerabilities	OWASP scan

---

71. Error Handling

Error	User Message (EN)	User Message (TA)	Recovery
Incorrect login	“Email or password incorrect. Try again.”	“மின்னஞ்சல் அல்லது கடவுச்சொல் தவறு. மீண்டும் முயற்சிக்கவும்.”	Retry; forgot password link
Expired session	“Your session expired. Please sign in again.”	“உங்கள் அமர்வு காலாவதியானது. மீண்டும் உள்நுழையவும்.”	Redirect to login
Lost network	“You’re offline. Some features unavailable.”	“நீங்கள் ஆஃப்லைனில் உள்ளீர்கள்.”	Retry; offline queue
Audio unavailable	“Audio can’t play right now.”	“ஆடியோ இப்போது இயங்காது.”	Retry; text fallback
Microphone denied	“Microphone access needed for speaking.”	“பேச மைக்ரோஃபோன் தேவை.”	Settings link; skip
AI timeout	“Tamil helper is busy. Try again.”	“தமிழ் உதவியாளர் பிஸியாக உள்ளார்.”	Retry; fallback content
Invalid AI response	“Something went wrong. Let’s try another way.”	“ஏதோ தவறு நடந்தது.”	Fallback to static content
DB write failure	“Progress couldn’t save. Retrying…”	“முன்னேற்றம் சேமிக்க முடியவில்லை.”	Retry with backoff
Duplicate completion	(Silent — idempotent)	(Silent)	Idempotency key
Browser refresh mid-lesson	(Resume prompt) “Continue where you left off?”	“தொடரவா?”	Resume from last item
Missing translation	(Fallback to English)	(Fallback to English)	Dev warning
Corrupted lesson data	“This lesson has an issue. Skipping…”	“இந்த பாடத்தில் சிக்கல்.”	Skip; report

Parent not authorised	“You don’t have access to this profile.”	“உங்களுக்கு அணுகல் இல்லை.”	Redirect to own dashboard
Account deletion	“Your account and data will be permanently deleted.”	“உங்கள் கணக்கு நிரந்தரமாக நீக்கப்படும்.”	Confirm dialog; cascade delete

---

72. Edge Cases

Edge Case	Handling
User changes device mid-lesson	Progress synced via Supabase; resume on new device
User loses streak due to technical error	Streak freeze applied; no penalty
Slow/low-memory device	Lazy loading; reduced animations; audio-only mode
Unsupported language selection	Fallback to English; log unsupported locale
Tamil text overflow	Text wrapping with proper line-height; no clipping
Combining character input	NFC normalization; correct rendering
Duplicate lesson submission	Idempotency key; only one record
Offline lesson completion	Queued locally; synced when online
Multiple children under one parent	Parent can switch between child profiles
Child turns 13	Account automatically transitions to standard privacy settings

---

73. Integrations

Integration	Purpose	Priority	Notes
Supabase	Backend (DB, Auth, Storage, RLS)	P0	Core infrastructure
Vercel / Netlify	Hosting	P0	Free tier sufficient
Sentry	Error monitoring	P1	Free tier
AI Provider API	Conversational practice, hints	P1	Edge Function only
Cloud TTS	Dynamic audio	P1	Fallback for prerecorded
Web Speech API	Speech recognition (optional)	P2	Browser support varies

---

74. Testing Strategy

Test Pyramid

Level	Coverage Target	Tools	Frequency
Unit	80% of utility functions	Vitest	Every commit
Component	70% of UI components	Vitest + Testing Library	Every PR
Integration	Critical user flows	Playwright	Daily
E2E	Onboarding → lesson → progress	Playwright	Pre-deploy
Database	RLS policies, constraints	Supabase test runner	Every migration
Localization	All translation keys exist	Custom script	Every PR
Accessibility	WCAG 2.1 AA	axe-core	Weekly
Performance	Load time, interaction	Lighthouse CI	Weekly
Security	OWASP Top 10	Manual + tools	Pre-launch

---

75. Test Cases

ID	Test	Preconditions	Steps	Expected	Severity	Priority
TC-001	Language	User logged in	Tap language	All UI strings change to	High	P0

	switch		selector → select Tamil	Tamil		
TC-002	Lesson resume	Lesson in progress	Close lesson → reopen	Resumes at last item	High	P0
TC-003	Correct answer	Lesson active	Submit correct answer	Positive feedback + XP awarded	Medium	P0
TC-004	Incorrect answer	Lesson active	Submit incorrect answer	Hint shown; retry allowed	Medium	P0
TC-005	Audio playback	Lesson with audio	Tap play	Audio plays within 2s	High	P0
TC-006	Progress persist	Lesson complete	Refresh page	Progress unchanged	High	P0
TC-007	RLS isolation	Two users	User A queries User B data	No results	Critical	P0
TC-008	Tamil rendering	Any screen	Display Tamil text	Correct ligatures; no clipping	High	P0
TC-009	Child consent	Under-13 signup	Attempt full access	Blocked until parental consent	Critical	P0
TC-010	Streak calculation	Daily activity	Complete lesson on consecutive days	Streak increments	Medium	P0
TC-011	Tanglish reduction	Stage A	Achieve 80% accuracy	Advance to Stage B	High	P1
TC-012	Offline handling	No network	Attempt lesson	Offline message; queued	Medium	P1
TC-013	Parent dashboard	Parent linked	View child progress	Data visible; read-only	Medium	P1
TC-014	AI hint	Lesson with hint	Request hint	Contextual hint shown	Low	P1
TC-015	Account deletion	User logged in	Request deletion	All data deleted within 30 days	Critical	P0

---

76. AI Evaluation

Evaluation Criteria for AI Features (P1)

Feature	Metric	Target	Method
Hint generation	Relevance (1–5)	>4.0	Human review of 50 samples
Hint generation	Safety (pass/fail)	100% pass	Content filter + human review
Conversational practice	Response appropriateness	>80%	Human review
Conversational practice	Grammar accuracy	>85%	Native speaker review
Adaptive revision	Precision of recommendations	>75%	A/B test
Literary explanation	Source accuracy	100%	Citation verification
Literary explanation	Hallucination rate	0%	Manual review of 100 outputs

Critical: AI is never used as the sole source of truth for foundational Tamil content.

---

77. Accessibility Evaluation

Automated + Manual Testing

Check	Tool	Frequency
Color contrast	axe-core, Stark	Every component
Keyboard navigation	Manual	Every screen

Screen reader (NVDA)	Manual	Weekly
Screen reader (VoiceOver)	Manual	Weekly
Reduced motion	Manual	Every animation
Tamil Unicode rendering	Visual regression	Every content update
Text resizing (200%)	Manual	Monthly

---

78. Performance Evaluation

Metric	Target	Tool	Frequency
LCP (Largest Contentful Paint)	<2.5s	Lighthouse	Weekly
FID (First Input Delay)	<100ms	Lighthouse	Weekly
CLS (Cumulative Layout Shift)	<0.1	Lighthouse	Weekly
Bundle size (initial)	<200KB gzipped	Vite build	Every PR
Audio load time	<2s	Network panel	Every audio asset
Tamil text render time	<500ms	Performance API	Every content update

---

79. Deployment

Environments

Environment	Purpose	URL	Database
Development	Local development	localhost:5173	Supabase local
Staging	Pre-production testing	staging.tamilpalam.vercel.app	Supabase staging project
Production	Live users	tamilpalam.vercel.app	Supabase production project

Deployment Pipeline

```
GitHub Push → Vercel Build → Preview Deploy → Manual Approval → Production Deploy
```

Step	Tool	Command
Build	Vite	`npm run build`
Test	Vitest + Playwright	`npm test`
Deploy	Vercel CLI / Git integration	Auto on main branch
Database migrations	Supabase CLI	`supabase db push`

Environment Variables

Variable	Scope	Secret	Description
`VITE_SUPABASE_URL`	Client	No	Supabase project URL
`VITE_SUPABASE_ANON_KEY`	Client	No	Supabase anon key
`SUPABASE_SERVICE_ROLE_KEY`	Server (Edge Function)	Yes	Admin access
`AI_API_KEY`	Server (Edge Function)	Yes	AI provider key
`SENTRY_DSN`	Client	No	Error monitoring

---

80. Monitoring

What	Tool	Alert Threshold
Uptime	Vercel / UptimeRobot	>99.5%
Error rate	Sentry	>1% of sessions
API latency	Supabase Dashboard	P95 >2s

API latency	Supabase Dashboard	p95 >2s
Database size	Supabase Dashboard	>80% of plan
Storage usage	Supabase Dashboard	>80% of plan
Auth failures	Supabase Dashboard	>10% of attempts

---

81. Backup and Recovery

Data	Backup Frequency	Retention	Recovery Time
PostgreSQL	Daily (Supabase)	7 days (free), 30 days (paid)	<1 hour
Storage (audio)	Weekly export	30 days	<4 hours
Code	Git (GitHub)	Permanent	<15 min
Environment config	Documented in secure vault	Permanent	<30 min

Recovery procedure: 1. Identify failure scope. 2. Restore database from most recent backup. 3. Verify data integrity. 4. Restore storage assets if affected. 5. Redeploy application. 6. Notify affected users.

---

82. Development Environments

Local Setup

```
# Prerequisites
Node.js 20+
npm 10+
Supabase CLI

# Steps
git clone https://github.com/team/tamilpalam.git
cd tamilpalam
npm install
cp .env.example .env.local
# Fill in Supabase credentials
supabase start  # Local Supabase
npm run dev     # Start Vite dev server
```

Required Tools

Tool	Version	Purpose
Node.js	20+	Runtime
npm	10+	Package manager
Supabase CLI	Latest	Local DB
Git	Latest	Version control
VS Code / Cursor	Latest	IDE

---

83. Repository Organisation

```
tamilpalam/
├── .github/
│   └── workflows/          # CI/CD
├── public/
│   ├── audio/              # Static audio fallbacks
│   └── images/             # Static images
├── src/
│   ├── app/
│   ├── components/
│   ├── hooks/
│   ├── lib/
│   ├── locales/
│   ├── stores/
│   ├── styles/
│   └── types/
├── supabase/
│   ├── migrations/         # SQL migrations
│   ├── functions/          # Edge Functions
│   └── seed.sql            # Seed data
├── tests/
│   ├── unit/
│   ├── component/
│   └── e2e/
├── docs/                   # PRD, architecture docs
├── .env.example
├── package.json
├── tsconfig.json
├── vite.config.ts
├── tailwind.config.js
└── README.md
```

---

84. Coding Conventions

Convention	Rule	Example
File naming	kebab-case for files, PascalCase for components	`lesson-player.tsx`, `LessonPlayer`
Component structure	One component per file	—
TypeScript	Strict mode; no `any`	`interface LessonProps`
Imports	Absolute imports with `@/` alias	`import { Button } from '@/components/ui/Button'`
State	Zustand for global; React state for local	—
CSS	Tailwind utility classes; no custom CSS unless necessary	`className="flex items-center gap-4"`
i18n	No hardcoded strings; use `t('key')`	`t('lesson.playAudio')`
Commits	Conventional commits	`feat: add language selector`
Branches	`feature/`, `fix/`, `chore/`	`feature/tanglish-reduction`

---

85. Team Responsibilities

Role	Primary Responsibilities	Secondary
Frontend Lead	UI components, routing, state management, responsive design	Visual design
Backend Lead	Supabase setup, RLS policies, Edge Functions, database migrations	API design
Curriculum Lead	Lesson content, exercises, vocabulary lists, Tanglish rules	Content review coordination
Integration/QA Lead	Testing, deployment, monitoring, audio assets, demo preparation	Documentation

---

86. Fifteen-Hour Execution Plan

Hour-by-Hour

Hour	Frontend Lead	Backend Lead	Curriculum Lead	QA/Integration Lead
0–1	Project setup, routing, design tokens	Supabase project, schema migration	Level 1 content outline	Repo setup, CI config
1–2	Auth UI, language selector	Auth setup, RLS policies	Level 1 letters + audio	Audio asset collection
2–3	Dashboard skeleton	Profiles table, preferences	Level 1 vocabulary	Test plan drafting
3–4	Lesson player shell	Lessons, lesson_items tables	Level 2 content	Unit tests for utils
4–5	Letter learning screen	Seed data for Levels 1–2	Level 2 audio	Component tests
5–6	Vocabulary screen	Progress tables, RLS	Level 3 content	E2E test: onboarding
6–7	Quiz/assessment screen	Achievements, XP logic	Level 3 audio	E2E test: lesson flow
7–8	Progress dashboard	Daily activity, streak logic	Tanglish reduction rules	Accessibility audit
8–9	Progress charts, badges	Revision queue logic	Level 4 content	Performance test
9–10	Tanglish reduction UI	Mastery calculation	Level 4 audio	Cross-browser test
10–11	Parent dashboard (P1)	Edge Function: AI hint (P1)	Content QA	Bug fixes
11–12	Polish, animations	Deployment config	Demo script	Demo rehearsal
12–13	Demo prep	Monitoring setup	Backup content	Final testing
13–14	Buffer	Buffer	Buffer	Buffer
14–15	**Demo**	**Demo**	**Demo**	**Demo**

Critical Path

```
Supabase schema → Auth → Dashboard → Lesson player → Progress → Demo
```

Feature Cut List (If Behind Schedule)

1. Cut parent dashboard (P1)
2. Cut AI features (P1)
3. Cut advanced animations
4. Cut Levels 5–7 content (keep Levels 0–4)
5. Cut speaking practice (keep listening + reading)
6. **Never cut:** Auth, lesson player, progress persistence, language switching, Tanglish reduction demo

Demo Strategy

Demo flow (5 minutes): 1. Landing page → language toggle (30s) 2. Sign up → onboarding (30s) 3. Assessment → level recommendation (30s) 4. Complete a letter lesson with audio (60s) 5. Show Tanglish reduction progression (60s) 6. View dashboard with progress, streak, badges (60s) 7. Show parent dashboard (30s)

Backup demo: Pre-recorded video + deployed staging URL + local build.

---

87. Task Dependency Map

```
flowchart TD
    A[Supabase Schema] --> B[Auth Setup]
    A --> C[Lesson Tables]
    B --> D[Dashboard]
    C --> E[Lesson Player]
    D --> F[Progress Display]
    E --> G[Quiz Logic]
    G --> H[Mastery Calculation]
    H --> I[Tanglish Reduction]
    C --> J[Seed Content]
    J --> E
    B --> K[RLS Policies]
    K --> D
    K --> F
```

---

88. Definition of Done

A feature is “done” when:

- [ ] Code reviewed by at least one other team member
- [ ] Unit tests written and passing
- [ ] Component test (if UI) passing
- [ ] E2E test covering happy path
- [ ] Accessibility check (keyboard, screen reader)
- [ ] Tamil rendering verified
- [ ] Translation keys added for both languages
- [ ] Responsive on mobile, tablet, desktop
- [ ] No console errors
- [ ] Deployed to staging
- [ ] Demo script updated

---

89. Demo Strategy

Element	Detail
Duration	5 minutes
Narrative	“Aarav learns Tamil” — follow a child from sign-up to first achievement
Key moments	Language toggle, audio playback, Tanglish reduction, progress celebration
Backup	Pre-recorded video, staging URL, local build on presenter laptop
Judges’ Q&A prep	Privacy compliance, progressive reduction algorithm, architecture choices

---

90. Acceptance Criteria

Feature	Acceptance Criteria	Priority
Language switching	All UI strings change; persists; lesson unaffected	P0
Onboarding	Completes <2 min; collects only display name + DOB	P0
Assessment	10 questions; recommends level; can override	P0
Lesson playback	Audio <2s load; controls responsive	P0
Progress persistence	Saved after each item; survives refresh	P0
Tamil rendering	No clipping, correct ligatures in all browsers	P0
Gamification	Streak, XP, badges visible and accurate	P0
Tanglish reduction	Stage advancement based on accuracy	P1
Child safety	Under-13 blocked until parental consent	P0
RLS isolation	Users cannot access other users’ data	P0

---

91. Risk Register

Risk	Likelihood	Impact	Detection	Mitigation	Fallback	Owner	Residual
Incorrect Tamil content	Medium	High	Reviewer check	Native speaker review	Content freeze; correction sprint	Curriculum Lead	Low
Poor transliteration	Medium	Medium	Learner confusion	Multiple alternatives shown	Audio as authority	Curriculum Lead	Low
Tanglish dependence	Medium	High	Mastery data stagnant	Progressive reduction enforced	Manual stage adjustment	Product Lead	Medium
Inaccurate AI feedback	Medium	High	User reports	Constrained prompts; human review	Disable AI feature	Backend Lead	Medium
Pronunciation	High	Medium	WER testing	No phoneme claims;	Skip AI	Backend	Low

assessment unreliable				self-assessment	pronunciation	Lead	
Children lose interest	Medium	High	Retention metrics	Gamification; short sessions	Content refresh	Product Lead	Medium
Scope creep	High	High	Timeline slip	Strict P0 focus	Cut P1/P2 features	Product Lead	Medium
Insufficient content	Medium	High	Lesson completion low	Seed Levels 1–4 only	Generate with review	Curriculum Lead	Low
Animation performance	Medium	Medium	Device testing	CSS-first; lazy 3D	Disable animations	Frontend Lead	Low
Network reliability	Medium	Medium	Offline testing	Offline queue; sync	Text-only mode	Backend Lead	Low
Privacy non-compliance	Low	Critical	Legal review	COPPA/GDPR-K checklist	Disable data collection	Product Lead	Low
Service outage	Low	High	Monitoring	Multi-provider fallback	Static demo	Backend Lead	Low
Copyright issues	Medium	High	Content review	Original content; licensed audio	Remove contested content	Curriculum Lead	Low
Incomplete localization	Medium	Medium	Missing key script	Automated check	English fallback	Frontend Lead	Low
Hackathon integration failure	Medium	High	Demo rehearsal	Integration early; contracts	Backup demo	QA Lead	Medium

92. Assumptions

1. The hackathon team has 4 members with basic web development skills.
2. Supabase free tier is sufficient for MVP.
3. Prerecorded audio can be sourced from team members or free libraries.
4. Tamil content can be reviewed by a native speaker on the team or a volunteer.
5. The hackathon provides internet access and deployment platform access.
6. Judges evaluate based on working demo, not production readiness.
7. English-speaking children abroad are the primary audience.
8. Tanglish is a familiar concept to the target audience's families.
9. Progressive Tanglish reduction is pedagogically sound (requiring validation).
10. AI features are optional enhancements, not core requirements.

93. Constraints

Constraint	Impact	Mitigation
15-hour deadline	Limits feature scope	Strict P0 prioritisation
4-person team	Limited parallel work	Clear role separation
No budget	Free tiers only	Supabase/Vercel free tier
No professional audio	Quality risk	Team recordings + free assets
No formal user testing	Assumptions unvalidated	Post-hackathon pilot plan
Browser compatibility	Tamil rendering varies	Extensive testing
Child privacy compliance	Limits data collection	Minimal data design

94. Open Questions

#	Question	Owner	Deadline	Impact if Unresolved
1	Is a native Tamil speaker available for content review?	Product Lead	Hour 1	Content quality risk

2	Can we source free Tamil audio assets?	Curriculum Lead	Hour 2	Audio quality risk
3	What is the exact COPPA consent method we can implement in 15 hours?	Product Lead	Hour 3	Child safety compliance
4	Is AI API access available within the hackathon?	Backend Lead	Hour 2	P1 feature scope
5	What is the deployment platform?	QA Lead	Hour 1	Deployment timeline
6	Can we validate Tamil rendering on all target browsers?	Frontend Lead	Hour 4	UX quality
7	What is the demo narrative?	Product Lead	Hour 10	Demo impact

---

95. Decision Log

Decision	Date	Rationale	Alternatives Considered
Use Supabase over Firebase	Day 0	PostgreSQL + RLS > NoSQL	Firebase (no RLS)
Use React + Vite	Day 0	Speed, TypeScript, AI tools	Next.js (overkill), Vue (less tooling)
No custom backend server	Day 0	Supabase client + RLS sufficient	Express (unnecessary)
P0 scope: Levels 0–4 only	Day 0	15-hour constraint	Full curriculum (impossible)
No AI in MVP	Day 0	Reliability, scope	AI hints (P1)
Prerecorded audio only	Day 0	Quality, offline	TTS (lower quality)
No leaderboards	Day 0	Child safety	Leaderboards (exposure risk)
English/Tamil full localization	Day 0	Core requirement	English-only (incomplete)

---

96. Future Roadmap

Stage 1 — Hackathon MVP (Now)

User Value: Working Tamil learning demo with core features. **Features:** P0 features as defined. **Dependencies:** Supabase, Vercel, audio assets. **Success Metrics:** Working demo; all P0 acceptance criteria met. **Risks:** Timeline; content quality.

Stage 2 — Improved Learning Experience (Month 1–3)

User Value: Polished, production-ready Tamil learning. **Features:** Progressive Tanglish Reduction fully implemented; personalized revision; adaptive recommendations; cultural storytelling; more content (Levels 5–7). **Dependencies:** User testing; content expansion. **Success Metrics:** 40% weekly return; 70% reach Stage C. **Risks:** Content velocity; engagement.

Stage 3 — Parent and Teacher Support (Month 3–6)

User Value: Parents and teachers can track and support learning. **Features:** Parent dashboard; teacher assignments; progress reports; email summaries. **Dependencies:** Parent-child linking; consent management. **Success Metrics:** 40% parent engagement; 20% teacher adoption. **Risks:** Privacy complexity; adoption.

Stage 4 — Advanced Speech and Personalization (Month 6–12)

User Value: Pronunciation feedback; truly adaptive learning. **Features:** Speech recognition (optional); phoneme-level assessment (if validated); advanced personalization. **Dependencies:** Tamil speech API availability. **Success Metrics:** 50% speaking participation; measurable pronunciation improvement. **Risks:** API accuracy; cost.

Stage 5 — Broader Curriculum (Year 2)

User Value: Full Tamil literacy from beginner to intermediate. **Features:** Complete Levels 0–10; advanced reading; writing practice; dialect awareness. **Dependencies:** Content team; reviewer network. **Success Metrics:** 60% completion to Level 6. **Risks:** Content scaling; quality maintenance.

Stage 6 — Tamil Research Assistant (Year 2+)

User Value: Literary Tamil understanding; Thirukkural study. **Features:** Passage explanation; source verification; uncertainty communication; hallucination prevention. **Dependencies:** Licensed content; RAG infrastructure. **Success Metrics:** 100% source accuracy; zero hallucinations. **Risks:** Licensing; accuracy.

---

97. Tamil Research Assistant Specification (Future Module)

Workflow

1. User selects or enters Tamil literary passage.
2. User selects explanation language (English/Tamil).
3. System identifies available source (Thirukkural, classical literature, proverbs).
4. System presents original passage separately from explanation.
5. System provides beginner-friendly explanation.
6. Tanglish assistance offered for relevant words.
7. Source details and uncertainty displayed.
8. If authenticity cannot be established, system says so.

Content Sources

Source	Licensing Status	Verification Required
Thirukkural	Public domain (ancient text)	Standard text confirmation
Classical literature	Varies by work	Licensing check
Proverbs	Oral tradition	Source attribution
Historical passages	Varies	Archive verification

Safety Requirements

- **Never invent quotations.**
- **Never claim unverified information is authentic.**
- **Always distinguish original text from explanation.**
- **Always cite sources where available.**
- **Always communicate uncertainty.**
- **Never present AI interpretation as scholarly consensus.**

Evaluation

Metric	Target	Method
Source accuracy	100%	Manual verification
Hallucination rate	0%	100-output review
Uncertainty communication	100%	Checklist review
User comprehension	>80%	Post-explanation quiz

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98. Content Sourcing and Licensing

Content Type	Source	License	Attribution Required
Tamil letters	Unicode standard	Open	No
Vocabulary	Original creation	Team-owned	No
Sentences	Original creation	Team-owned	No
Stories (original)	Team creation	Team-owned	No
Stories (folk tales)	Public domain	Public domain	Yes (source)
Audio (team)	Team recordings	Team-owned	No
Audio (licensed)	Free libraries	Check license	Yes

Images	Original / licensed	Check license	Yes

Critical: Do not use copyrighted Tamil textbooks, proprietary audio, or unlicensed literary content.

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99. Budget and Operating-Cost Estimates

Item	Free Tier	Paid Estimate	Notes
Supabase	500MB DB, 1GB storage	$25/month (Pro)	Sufficient for MVP
Vercel	100GB bandwidth	$20/month (Pro)	Free tier sufficient
Domain	—	$12/year	`.app` or `.com`
AI API	Limited free	$10–50/month	P1 only
Audio recording	Team	$0–100 one-time	If professional needed
Sentry	5K events/month	$26/month	Free tier sufficient
Total (MVP)	**$0**	—	Free tier sufficient
Total (Year 1, paid)	—	**~$800**	With AI + Pro tiers

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100. Maintenance Strategy

Task	Frequency	Owner	Tool
Dependency updates	Monthly	Frontend Lead	Dependabot
Security patches	As needed	Backend Lead	npm audit
Content corrections	As reported	Curriculum Lead	CMS
Database backup verification	Monthly	Backend Lead	Supabase
Performance review	Monthly	QA Lead	Lighthouse
User feedback review	Weekly	Product Lead	In-app reports
Accessibility audit	Quarterly	QA Lead	axe-core
Privacy policy review	Annually	Product Lead	Legal

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101. Product Validation Plan

Phase	Method	Sample	Duration	Success Criteria
Alpha	Team testing	4 users	Week 1	No critical bugs
Beta	Friends/family	10 users	Week 2–3	70% complete onboarding
Pilot	Tamil school partnership	20 children	Month 2	50% weekly return
Validation	Formal study	100 learners	Month 3–6	Measurable script improvement

Critical: Do not claim educational efficacy until validated by a controlled study.

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102. User Research Plan

Research Question	Method	Sample	Timeline
Do children find the interface intuitive?	Moderated usability test	5 children	Week 2
Does Tanglish reduction feel natural?	Diary study	10 learners	Week 4
What motivates continued use?	Survey + interviews	20 users	Month 2
Do parents find the dashboard useful?	Parent interviews	10 parents	Month 2
Are Tamil educators satisfied with content?	Expert review	3 educators	Month 1

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103. Pilot Deployment Plan

Step	Detail	Timeline
Partner identification	Tamil school / community organisation	Week 1
Consent collection	Parental consent for all participants	Week 2
Onboarding session	In-person or video call setup	Week 3
Pilot period	4 weeks of regular use	Weeks 3–7
Data collection	Usage + quiz data (anonymized)	Continuous
Exit interviews	Learners + parents	Week 7
Report	Findings + recommendations	Week 8

---

104. Growth and Adoption Strategy

Channel	Tactics	Target
Tamil community organisations	Partnerships with Tamil schools	5 schools in Year 1
Parent networks	Referral programme (non-monetary)	20% referral rate
Social media	Tamil parenting groups	Organic reach
Content marketing	Blog posts on Tamil learning	SEO for “learn Tamil for kids”
App stores	PWA distribution	—
Teacher ambassadors	Free classroom accounts	10 teachers

Child safety in marketing: No targeting of children with ads. Marketing directed at parents and educators.

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105. Final Requirements Traceability Matrix

Requirement	Source	Priority	Test Case	Status
FR-001 Language switching	P10 problem	P0	TC-001	Planned
FR-002 Onboarding	Section 9	P0	TC-001	Planned
FR-003 Assessment	Section 9	P0	—	Planned
FR-004 Audio playback	Section 14	P0	TC-005	Planned
FR-005 Progress persistence	Section 9	P0	TC-006	Planned
FR-006 RLS isolation	Section 19	P0	TC-007	Planned
FR-007 Tamil rendering	Section 8	P0	TC-008	Planned
FR-008 Child consent	Section 19	P0	TC-009	Planned
FR-009 Tanglish reduction	Section 11	P1	TC-011	Planned
FR-010 AI hints	Section 13	P1	TC-014	Planned
FR-011 Parent dashboard	Section 16	P1	TC-013	Planned
FR-012 Research assistant	Section 15	P2	—	Future

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106. Glossary

Term	Definition
Tanglish	Tamil language written using English/Latin script
Uyir	Tamil independent vowels (12 letters)
Mei	Tamil consonants (18 letters)
Uyirmei	Combined consonant + vowel forms (216 forms)
Virama/Pulli	Diacritic that suppresses inherent vowel ◌் ()

Retroflex	Consonants pronounced with tongue curled back (ட, ண, ள, ழ, ற)
RLS	Row Level Security — PostgreSQL feature for row-level access control
CEFR	Common European Framework of Reference for Languages
COPPA	Children's Online Privacy Protection Act (US)
GDPR-K	General Data Protection Regulation — children's provisions (EU)
NFC	Unicode Normalization Form C — canonical composition
WER	Word Error Rate — speech recognition accuracy metric
SRS	Spaced Repetition System
LCP	Largest Contentful Paint — performance metric
PWA	Progressive Web App

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107. Appendices

Appendix A: Tamil Unicode Quick Reference

Category	Range	Count
Independent vowels	U+0B85–U+0B94	12
Consonants	U+0B95–U+0BB9	18
Vowel signs	U+0BBE–U+0BCC	13
Virama	U+0BCD	1
Numerals	U+0BE6–U+0BEF	10

Appendix B: Sample Seed Data

```
        -- Sample learning level
        INSERT INTO learning_levels (level_number, title_en, title_ta, description_en)
        VALUES (1, 'Tamil Letters and Sounds', 'தமிழ் எழுத்துக்கள்', 'Learn vowels and consonants');

        -- Sample lesson
        INSERT INTO lessons (level_id, title_en, title_ta, sequence_order, lesson_type)
        VALUES ((SELECT id FROM learning_levels WHERE level_number = 1),
                'Vowels: அ ஆ இ ஈ', 'உயிர் எழுத்துக்கள்', 1, 'letters');

        -- Sample letter
        INSERT INTO tamil_letters (letter, unicode_codepoint, letter_type, tanglish_primary, ipa)
        VALUES ('அ', 'U+0B85', 'vowel', 'a', '/a/');
```

Appendix C: RLS Policy Template

```
-- Standard user-owned table policy
ALTER TABLE {table_name} ENABLE ROW LEVEL SECURITY;

CREATE POLICY "Users can view own {table_name}"
ON {table_name} FOR SELECT
TO authenticated
USING (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));

CREATE POLICY "Users can insert own {table_name}"
ON {table_name} FOR INSERT
TO authenticated
WITH CHECK (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));

CREATE POLICY "Users can update own {table_name}"
ON {table_name} FOR UPDATE
TO authenticated
USING (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
))
WITH CHECK (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));

CREATE POLICY "Users can delete own {table_name}"
ON {table_name} FOR DELETE
TO authenticated
USING (profile_id IN (
  SELECT id FROM profiles WHERE user_id = auth.uid()
));
```

Appendix D: COPPA/GDPR-K Compliance Checklist

- [ ] Verifiable parental consent before data collection (under-13)
- [ ] Clear, plain-language privacy policy
- [ ] Separate children's privacy section
- [ ] Parental access to child data
- [ ] Parental deletion rights
- [ ] Data minimisation (collect only what's needed)
- [ ] No behavioural advertising
- [ ] No third-party data sharing without consent
- [ ] Data retention limits defined
- [ ] Voice data protection (24-hour deletion)
- [ ] No monetisation of child data
- [ ] Third-party SDK audit completed

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END OF PRD — TamilPalam v1.0

This document constitutes the complete Product Requirements Document for TamilPalam. All sections (1–108) are included. No part has been omitted or marked “to be continued.”